04 Electrochemistry

Introduction :

Electrochemistry deals with the study of electrical properties of solution of electrolytes and the inter-relation of chemical phenomenon and electrical energies.

Electric Conductors are of two types :

(i) Metallic conductors (ii) Electrolytic conductors or electrolytes.

(i) Metallic conductors :

The conductors which conduct electric current by movement of free electrons without undergoing any chemical change are known as metallic conductors.

eg. Metals : Cu, Ag, Fe, Al etc., non-metals : graphite and various alloys.

(ii) Electrolytic conductors :

Those substances whose aqueous solution conducts the electric current and which are decomposed by the passage of DC current are called electrolytes. In this case, conduction takes place by movement of ions.

Electrolytes also conduct electricity in fused state and undergo decomposition by passage the electric current.

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Strong electrolyte :

Electrolytes which are completely ionized in aqueous solution or in their molten state, are called strong electrolytes. Their aqueous solutions are strongly conducting.

Ex : All salts, strong acids and strong bases.

Weak electrolyte :

Electrolytes which are not completely ionized in aqueous solution are called **Weak electrolytes.** Their aqueous solutions are weakly conducting.

Ex: Organic acids CH_3COOH, HCN (Except : Alkyl sulphonic acids, RSO_3H)

Organic base : Amines, Aniline etc.

Note :

Ostwald's dilution law is only applicable for weak electrolytes according to which degree of dissociation(α) increases on dilution.

- For weak electrolyte : $\alpha << 1 \Rightarrow$ lesser ions $\Rightarrow$ weakly conducting
- For strong electrolyte : $\alpha = 1$ (always) $\Rightarrow$ more ions $\Rightarrow$ Highly conducting

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Difference between metallic and electrolytic conduction:

Metallic conduction	Electrolytic conduction
(i) Flow of electricity takes place due to free electrons without the decomposition of the substance.	Flow of electricity takes place by ions.
(ii) No transfer of matter takes place.	Transfer of matter takes place.
(iii) The resistance to the flow of current increases with the increase in temperature and hence the increase in temperature decreases the conduction.	The resistance to the flow of current decreases with the increase in temperature and hence increase in temperature increases the conduction.

Electrochemical cell :

It is a device which converts chemical energy into electrical energy or vice versa.

Electrochemical cells are of two types

Galvanic cell or Voltaic cell

- A spontaneous chemical reaction generates an electric current or EMF
- Chemical energy converted into electrical energy.

Electrolytic cell.

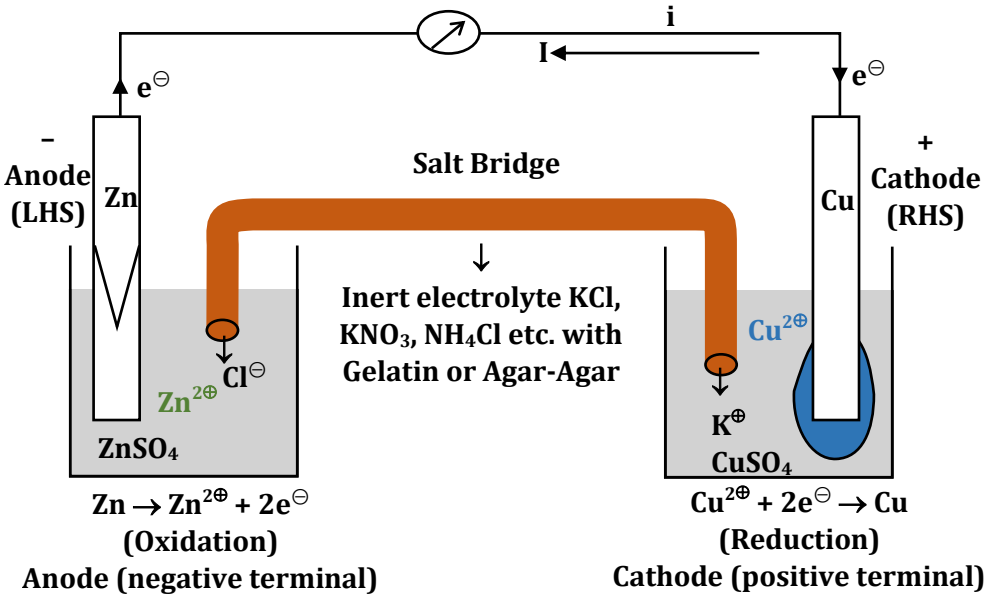
- An electric current drives a non-spontaneous reaction.
- Electrical energy converted into chemical energy.

Galvanic cell or Voltaic cell :

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- A cell in which the chemical energy is transformed into electrical energy.
- The chemical reaction occurring in a galvanic cell is a spontaneous redox reaction.

Galvanic cell is made up of two half cells i.e. anodic and cathodic. The cell reaction is of redox kind. Oxidation takes place at anode and reduction at cathode. It may be represented by Daniel cell which is a type of Galvanic cell. Zinc rod immersed in $ZnSO_4$ behaves as anode and copper rod immersed in $CuSO_4$ behaves as cathode.



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Oxidation takes place at anode :

$Zn(s) \longrightarrow Zn^{2\oplus} + 2e^{\ominus}$ (Loss of electron : Oxidation)

Reduction takes place at cathode :

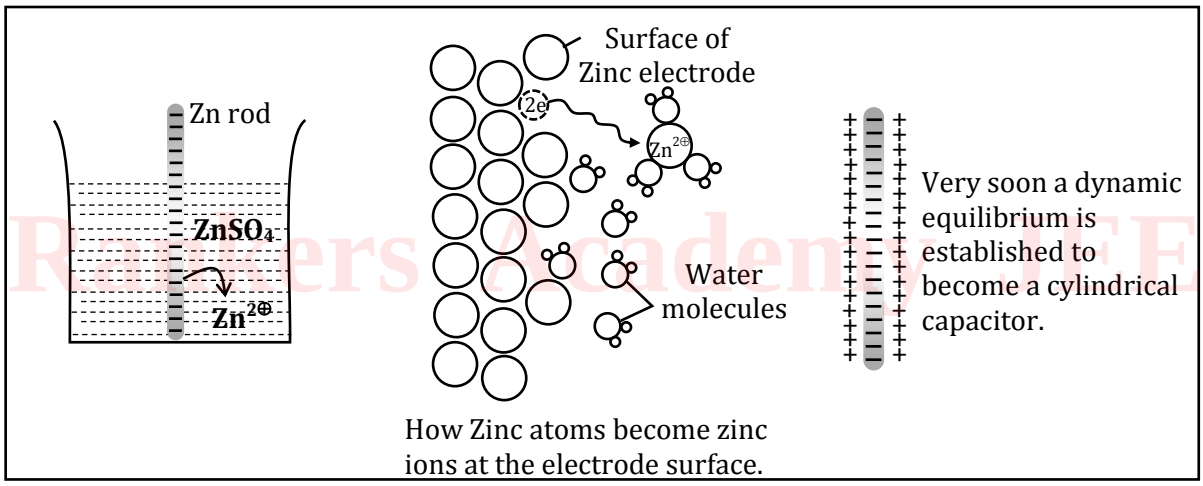
$Cu^{2\oplus} + 2e^{\ominus} \longrightarrow Cu(s)$ (Gain of electron : Reduction)

Construction of Cell :

- It has two half–cells, each having a beaker containing a metal strip that dips in its aqueous solution.
- The metal strips are called **electrodes** and are connected by a conducting wire.
- Two solutions are connected by a **salt bridge.**

Construction and working principle of Daniel cell :

I. Anode of Daniel cell : Zn rod placed in $ZnSO_4$ solution is found to have tendency to go into the solution phase as it is in contact with their ions or their salt solutions.



The Zn atom or metal atoms will move in the solution to form $Zn^{2\oplus}$. After some time following equilibrium will be established.

$Zn(s) \rightleftharpoons Zn^{2\oplus} + 2e^{\ominus}$

There will be accumulation of sufficient negative charge on the rod which will not allow extra zinc ions to move in the solution. i.e. solution will be saturated with $Zn^{2\oplus}$ ions.

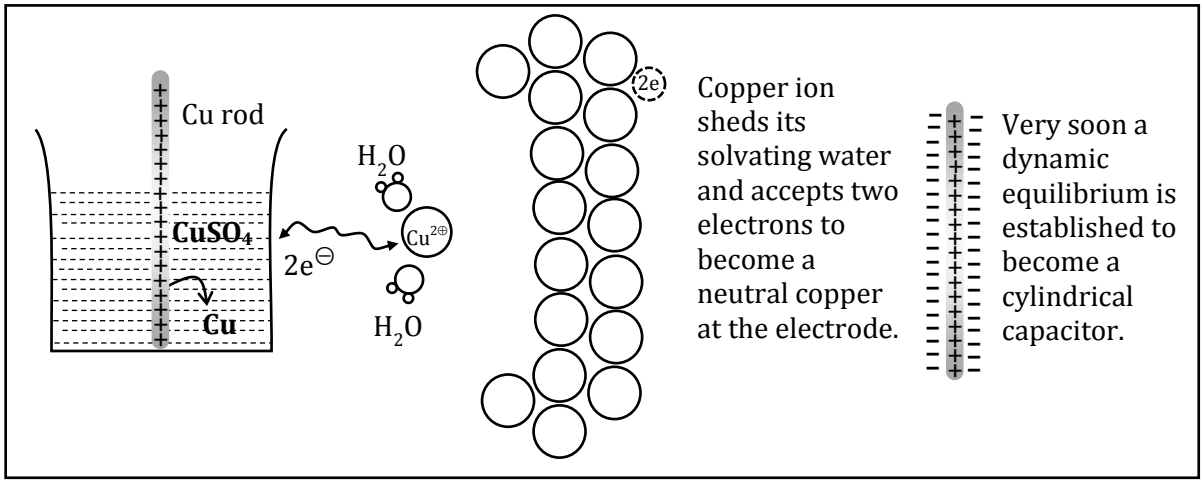
The extra positive charge of the solution will be more concentrated around the negatively charged rod. An electrical double layer is developed in the system and hence a potential difference is created between the rod and the solution which is known as electrode potential.

This particular electrode is known as anode :

- At anode oxidation will take place. (Release of electron).
- It is of negative polarity.
- The electrode oxidation potential is represented by $E_{Zn(s)/Zn^{2\oplus}(aq.)}$ & reduction potential by $E_{Zn^{2\oplus}(aq)/Zn(s)}$.

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II. Cathode of Daniel cell :



Cu, when placed in contact with their aqueous ions, the ions ($Cu^{2\oplus}$) from the solution will get deposited on the metal rod.

The following equilibrium will be established :

$Cu^{2\oplus}(aq) + 2e^{\ominus} \rightleftharpoons Cu(s)$

So, rod will have deficiency of electron (that is positively charged). Extra negative charge will surround this positively charged rod and form double layer. An electrical double layer is developed in the system and hence a potential difference is created between the rod and the solution which is known as electrode potential. This will be known as cathode.

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- At cathode, reduction will take place. (Gain of $e^{\ominus}$ will take place)
- Cathode acts as sink of electron.
- Positive polarity will be developed.
- Their electrode reduction potential can be represented by : $E_{Cu^{2\oplus}(aq)|Cu(s)}$.

- Anode : { Has a negative sign ; Is where oxidation occurs }
- Cathode : { Is where reduction occurs ; Has a positive sign }

Overall process :

$Zn(s) + Cu^{2\oplus}(aq) \longrightarrow Cu(s) + Zn^{2\oplus}(aq)$

In Galvanic cell like Daniel cell ; electrons flow from anode (zinc rod) to the cathode (copper rod) through external circuit ; zinc dissolves as $Zn^{2\oplus}$; $Cu^{2\oplus}$ ion in the cathode cell picks up two electron and is deposited at cathode.

Note :

The electrode potential will keep on decreasing with time as $Zn^{2\oplus}$ ions increase & $Cu^{2\oplus}$ ions decrease in solution therefore tendency of cell reaction decreases and cell attains equilibrium.

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Functions of Salt Bridge :

- A **salt bridge** is a U-shaped inverted tube that contains a gel permeated with an inert electrolyte.
- It connects the solution of two half-cell to complete the circuit.
- It minimises the liquid junction potential (the potential difference between the junction of two liquids)
- It maintains the electrical neutrality of the solution in order to give continuous flow or generation of current. The simultaneous electrical neutrality of the anodic oxidation chamber and cathodic reduction chamber is due to almost same mobility or velocity of $K^{\oplus}$ and $NO_3^{\ominus}$ ions taken into salt bridge.
- If the salt bridge is removed then voltage drops to zero & cell stops working.
- The ions of the inert electrolyte should not react with other ions in the solution and the ions are not oxidised or reduced at the electrodes.
- Generally, tube is filled with a paste of agar-agar powder with neutral electrolyte generally not common to anodic/cathodic compartment with porous plugs at each mouth of tube.
- It prevents mechanical mixing of two electrolytic solutions.

Liquid Junction Potential :

The potential difference which arises between two solutions when in contact with each other. Salt bridge removes effects of junction potential by providing appropriate migration of ions.

Characteristics of electrolyte used in salt bridge :

1. The electrolyte should be inert.
2. The cations and anions of the electrolyte used should be of almost equal ionic mobility.
3. Ions of electrolyte should not react with ions involved in cell reaction.

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Representation of galvanic cell (According to IUPAC) :

The anode is written on the LHS & cathode on the RHS.

we denote salt bridge by two vertical parallel lines (||), if used, in between anode & cathode.

Ex. $Pt(s)|H_2(g)|H^{\oplus}(aq.)|\ H^{\oplus}(aq.)|H_2(g)|Pt(s)$

Ex. Daniel cell : $Zn\ (s)\ |\ Zn^{2\oplus}(aq)\ ||\ Cu^{2\oplus}(aq.)\ |\ Cu\ (s)$

Ex. For cell reaction : $H_2(g) + Cu^{2\oplus}(aq) \longrightarrow 2H^{\oplus}(aq) + Cu(s)$

$Pt\ |\ H_2\ (g)\ |\ H^{\oplus}\ (aq)\ ||\ Cu^{2\oplus}(aq.)\ |\ Cu\ (s).$

Electrode potential :

When a strip of metal is brought in contact with the solution containing its own ions then the strip of metal gets positively charged or negatively charged and results into a potential being developed between the metallic strip and its solution which is known as electrode potential.

- **At anode :** $M \rightarrow M^{n\oplus} + ne^{\ominus}$ (Oxidation Potential)
- **At cathode :** $M^{n\oplus} + ne^{\ominus} \rightarrow M$ (Reduction Potential)
- The value of electrode potential depends upon :
  (i) the nature of electrode
  (ii) the concentration of solution
  (iii) the temperature

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Standard electrode potential (E°) :

If the concentration of ions is unity, temperature is 25°C (or any constant temperature) and pressure is 1 bar (standard conditions), the potential of the electrode is called **standard electrode potential.**

- The given value of electrode potential is regarded as reduction potential unless it is specifically mentioned that it is an oxidation potential.

➢ Electrode potential arises due to difference in charge distribution between the two "surfaces".
  - Metal electrode
  - Solvated ions surface

➢ This potential difference is the electrode potential for the couple.

➢ It is impossible to measure as it is difficult to know where to exactly put the wire for measuring the potential difference.

➢ Electrode potential is measured with respect to the reference electrode.

Different types of electrodes:

I. Metal - Metal ion electrode :

Cathode : $M^{n\oplus}(aq) \mid M(s)$

$M^{n\oplus} + ne^{\ominus} \longrightarrow M(s)$

$$E_{M^{n\oplus}/M} = E^0_{M^{n\oplus}/M} - \frac{0.059}{n}\log\frac{1}{[M^{n\oplus}]}$$

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II. Gas – Gas ion electrode :

Anode : $Pt(s) \mid H_2$ (P atm) $\mid H^{\oplus}$ (c M)

Cathode : $H^{\oplus}$(c M) $\mid H_2$ (P atm)$\mid$ Pt(s)

Cathodic reaction : $H^{\oplus}(aq) + e^{\ominus} \longrightarrow \frac{1}{2}H_2$ (P atm)

$$E = E° - 0.0591 \log\frac{P_{H_2}^{\frac{1}{2}}}{[H^{\oplus}]} = -0.0591\ pH \left[\because E^0_{H^{\oplus}/H_2} = 0 \& p_{H_2} = 1\,bar\right]$$

III. Oxidation - reduction electrode (or redox electrode) :

It has same metal (or element) in two different oxidation states in same solution.

Anode : $Pt(s) \mid Fe^{2\oplus}(aq), Fe^{3\oplus}(aq)$

Cathode : $Fe^{3\oplus}(aq), Fe^{2\oplus}(aq) \mid Pt(s)$

$Fe^{3\oplus} + e^{\ominus} \longrightarrow Fe^{2\oplus}$

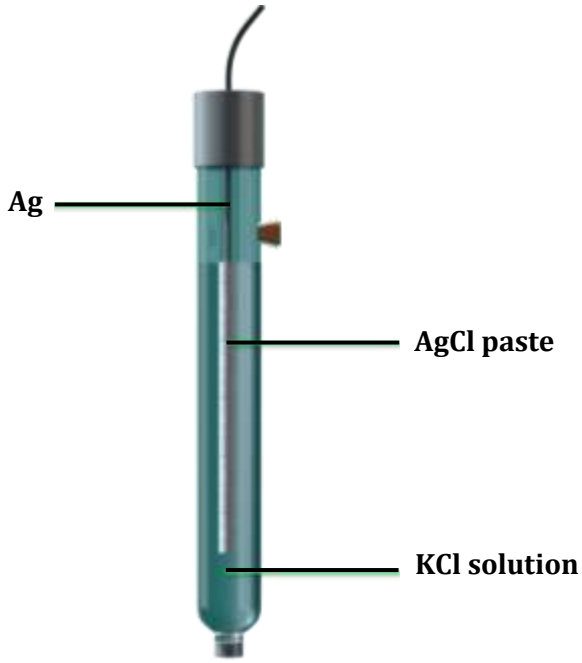
$$E = E° - 0.0591 \log\left[\frac{Fe^{2\oplus}}{Fe^{3\oplus}}\right]$$

Also, $Cr_2O_7^{2\ominus}(aq.), Cr^{3\oplus}(aq.), H^{\oplus} \mid Pt$

$Pt \mid Mn^{2\oplus}(aq.), MnO_4^{\ominus}(aq.), H^{\oplus}$

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IV. Metal - metal insoluble salt-salt anion electrode :



In this half-cell, a metal coated with its insoluble salt is in contact with a solution containing the anion of the insoluble salt. eg. Silver-Silver Chloride Half Cell:

This half-cell is represented as $Cl^{\ominus}|AgCl|Ag$. The equilibrium reaction that occurs at the electrode is

$AgCl(s) + e^{\ominus} \rightleftharpoons Ag(s) + Cl^{\ominus}(aq)$

$$E_{Cl^{\ominus}/AgCl/Ag} = E^{0}_{Cl^{\ominus}/AgCl/Ag} - \frac{0.0591}{1}\log\left[Cl^{\ominus}\right]$$

potential of such cells depends upon the concentration of anions. Such cells can be used as **Reference Electrode**.

To calculate standard potential of any other electrode a cell is coupled with standard hydrogen electrode (SHE) and it's potential is measured that gives the value of electrode potential of that electrode.

Ex.

Anode : Zinc electrode

Cathode : SHE

Cell : Zinc electrode || SHE

$E_{cell} = E_{H^{\oplus}/H_2(g)} - E^{o}_{Zn^{2\oplus}|Zn}$

$= 0.76$ V (at 298 K experimentally)

So, $E^{o}_{Zn^{2\oplus}|Zn} = -0.76$ V (SRP) $\Rightarrow E^{o}_{Zn|Zn^{2\oplus}(aq)} = 0.76$ V (SOP)

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Reference Electrode :

Absolute values of electrode potentials cannot be measured. Reference electrodes is an electrode used to measure the electrode potential of other electrodes.

(a) Standard Hydrogen Electrode (SHE) :

It consists of a platinum electrode over which H_2 gas (1 bar pressure) is bubbled and the electrode is immersed in a solution that is 1 M in $H^{\oplus}$ at any specified temperature.

$$2H^{\oplus} + 2e^{\ominus} \longrightarrow H_2(g)$$
(1 M) (1 atm)

The potential of this electrode at all temperature is taken as Zero volt. [IUPAC convention]

Calculation of electrode potential :

At Anode

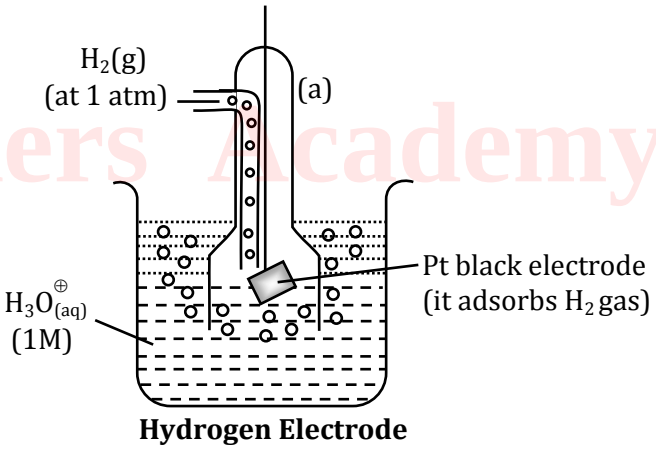
$H_2(g) \rightleftharpoons 2H^{\oplus} + 2e^{\ominus}$

- Oxidation potential O.P. = $E_{H_2(g)/H^{\oplus}(aq.)}$
- $E^0_{H_2(g)/H^{\oplus}(aq.)}$ = SOP = 0
- $E_{H_2/H^{\oplus}} \neq 0$

At Cathode

$2H^{\oplus} + 2e^{\ominus} \rightleftharpoons H_2(g)$

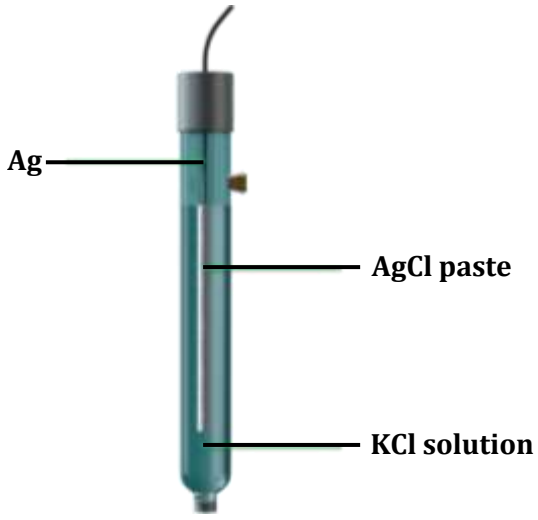
- Reduction Potential (R.P.) = $E_{H^{\oplus}/H_2(g)}$
- $E^0_{H^{\oplus}/H_2(g)}$ = SRP = 0
- $E_{H^{\oplus}/H_2} \neq 0$



Hydrogen Electrode

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(b) Ag-AgCl electrode :



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Anode:

$Ag(s)/AgCl(s)/Cl^{\ominus}(aq)$

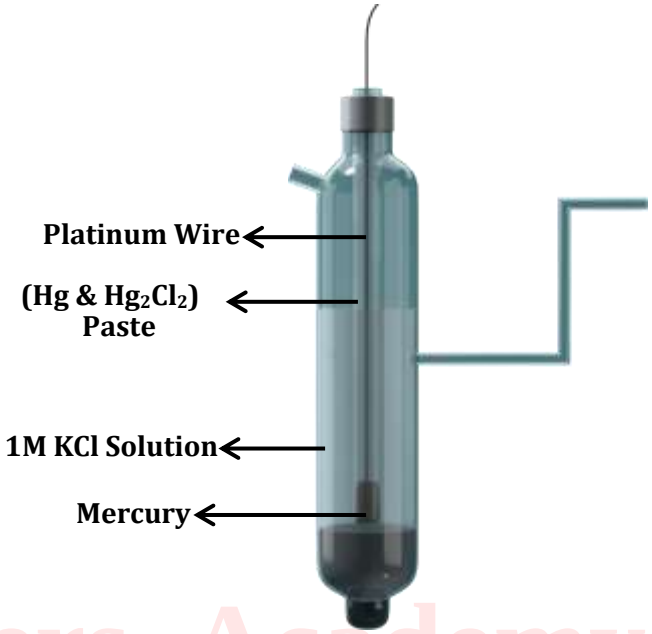
$Ag(s) + Cl^{\ominus}(aq) \rightarrow AgCl(s) + e^{\ominus}$

This half reaction takes place in 2 steps:

(i) $Ag(s) \rightarrow Ag^{\oplus}(aq) + e^{\ominus}$

(ii) $Ag^{\oplus}(aq) + Cl^{\ominus}(aq) \rightarrow AgCl(s)$

(c) Calomel electrode:



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Cathode:

$Cl^{\ominus}(C\ M) \mid Hg_2Cl_2(s) \mid Hg(\ell)|Pt(s)$

$Hg_2Cl_2(s) + 2e^{\ominus} \rightarrow 2Hg(\ell) + 2\ Cl^{\ominus}(aq)$

This half reaction takes place in 2 steps:

(i) $Hg_2Cl_2(s) \rightarrow Hg_2^{2\oplus}(aq) + 2Cl^{\ominus}$

(ii) $Hg_2^{2\oplus}(aq) + 2e^{\ominus} \rightarrow 2Hg(\ell)$

Cathode : $Cl^{\ominus}(c\ M) |Hg_2Cl_2(s) \mid Hg(l)| Pt(s)$

It is prepared by a Pt wire in contact with a paste of Hg and Hg_2Cl_2 present in a KCl solution.

Reaction $Hg_2Cl_2(s) + 2e^{\ominus} \longrightarrow 2Hg(l) + 2Cl^{\ominus}$; $E^0 = +0.27$ V

$$\Rightarrow \boxed{E_{Cl^{\ominus}/Hg_2Cl_2/Hg} = E^0_{Cl^{\ominus}/Hg_2Cl_2/Hg} - \frac{0.059}{2}\log[Cl^{\ominus}]^2}.$$

Electromotive force of cell or cell voltage :

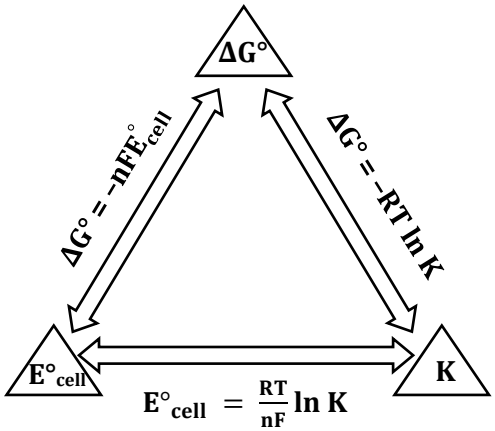
The difference in the electrode potentials of the two electrodes of the cell is termed as electro motive force [EMF] or cell voltage which causes current to flow.

$E_{cell} = E_{red}$ (cathode) $- E_{red}$ (anode) $= E_{oxi.}$(anode) $- E_{oxi.}$(cathode) $= E_{oxi.}$(anode) $+ E_{red}$(cathode)

During the chemical process, the reduction in Gibbs free energy is converted in the form of electrical energy.

$(\Delta G)_{T,P} = w_{Useful|max.} = -nFE_{cell.}$

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ΔG°	K	E^{o}_{cell}	Reactions at standard-state conditions
<0	>1	>0	Spontaneous
0	1	0	At equilibrium
>0	<1	<0	Non-spontaneous

Electrochemical series: Standard aqueous electrode potentials at 298 K

Electrode Reduction Reaction	Standard electrode Reduction potential E° (Volts)
$Li^{\oplus} + e \rightarrow Li$	– 3.05
$K^{\oplus} + e^{\ominus} \rightarrow K$	– 2.93
$Ba^{2\oplus} + 2e^{\ominus} \rightarrow Ba$	– 2.90
$Ca^{2\oplus} + 2e^{\ominus} \rightarrow Ca$	– 2.87
$Na^{\oplus} + e^{\ominus} \rightarrow Na$	– 2.71
$Mg^{2\oplus} + 2e^{\ominus} \rightarrow Mg$	– 2.37
$Al^{3\oplus} + 3e^{\ominus} \rightarrow Al$	– 1.66
$Mn^{2\oplus} + 2e^{\ominus} \rightarrow Mn$	– 1.18
$H_2O + e^{\ominus} \rightarrow \frac{1}{2} H_2 + OH$	– 0.83
$Zn^{2\oplus} + 2e^{\ominus} \rightarrow Zn$	– 0.76
$Cr^{3\oplus} + 3e^{\ominus} \rightarrow Cr$	– 0.74
$Fe^{2\oplus} + 2e^{\ominus} \rightarrow Fe$	– 0.44
$Cd^{2\oplus} + 2e^{\ominus} \rightarrow Cd$	– 0.40
$Ni^{2\oplus} + 2e^{\ominus} \rightarrow Ni$	– 0.25
$Sn^{2\oplus} + 2e^{\ominus} \rightarrow Sn$	– 0.14
$Pb^{2\oplus} + 2e^{\ominus} \rightarrow Pb$	– 0.13
$2D^{\oplus} + e^{\ominus} \rightarrow D_2$	– 0.01 V
$2H^{\oplus} + 2e^{\ominus} \rightarrow H_2$	0
$AgBr(s) + e^{\ominus} \rightarrow Ag(s) + Br^{\ominus}$	+0.09 V

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$AgCl + e^{\ominus} \rightarrow Ag + Cl^{\ominus}$	0.22 V
$Cu^{2\oplus} + 2e^{\ominus} \rightarrow Cu$	+ 0.34
$¼\ O_2 + ½\ H_2O + e^{\ominus} \rightarrow OH^{\ominus}$	+0.401 V
$I_2 + 2e^{\ominus} \rightarrow 2I^{\ominus}$	+ 0.54
$Hg_2^{2\oplus} + 2e \rightarrow 2Hg$	+ 0.79
$Ag^{\oplus} + e^{\ominus} \rightarrow Ag$	+ 0.80
$Hg^{2\oplus} + 2e^{\ominus} \rightarrow Hg$	+ 0.85
$Br_2 + 2e^{\ominus} \rightarrow 2Br^{\ominus}$	+ 1.08
$¼\ O_2 + H^{\oplus} + e^{\ominus} \rightarrow ½\ H_2O$	+1.23 V
$Cl_2 + 2e^{\ominus} \rightarrow 2Cl^{\ominus}$	+ 1.36
$Pt^{2\oplus} + 2e^{\ominus} \rightarrow Pt$	+ 1.20
$Au^{3\oplus} + 3e \rightarrow Au$	+ 1.50
$Au^{\oplus} + e \rightarrow Au$	+ 1.69
$S_2O_8^{2\ominus} + 2e^{\ominus} \rightarrow 2SO_4^{2\ominus}$	+2.0 V
$F_2 + 2e^{\ominus} \rightarrow 2F^{\ominus}$	+ 2.87

Application of electrochemical series :

(i) Activity of metal :

From electrochemical series, the activity of any metal can easily be determined. All the metals which are placed above hydrogen are stronger reducing agents & can easily evolve H_2 gas whereas metals lying below hydrogen are weaker reducing agents and cannot lose electrons to $H^{\oplus}$ ions & hence can't evolve H_2 gas. For e.g. Na, K, Zn etc. can easily evolve H_2 whereas Cu, Hg, Ag etc. do not have tendency to evolve H_2 gas.

$Na + H_2SO_4 \rightarrow Na_2SO_4 + H_2\uparrow$

$Na + H_2O \rightarrow NaOH + H_2\uparrow$

$Cu + H_2SO_4 \not\rightarrow CuSO_4 + H_2\uparrow$

(ii) Displacement reaction :

The active metal can easily displace less active metal from their aq. salt solution for e.g. Zn can replace $Cu^{2\oplus}$ from an aq. solution of $CuSO_4$. But Cu cannot displace $Zn^{2\oplus}$ from solution in the same way.

$2AgNO_3 + Cu \rightarrow Cu(NO_3)_2 + 2Ag,$

$CuSO_4 + Ag \not\rightarrow Ag_2SO_4 + Cu$

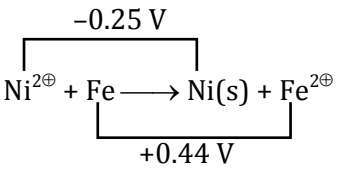
(iii) Feasibility of redox reaction :

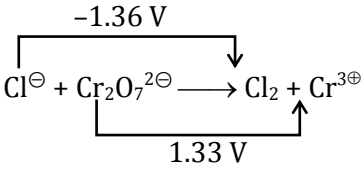
The feasibility of particular redox reaction can be easily found out from electrochemical series. The metal placed higher or have more reducing property can easily lose electrons to the metal ion present below in series, hence redox reaction become feasible i.e. cell will serve as source of electrical energy. For e.g. $NiSO_4$ solution cannot be placed in Fe vessel because, the redox reaction between them is feasible.

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Note :

If emf of the cell for redox reaction comes out to be positive, it suggests the redox reaction is spontaneous or feasible. Negative value indicate that redox reaction is not feasible.





(iv) Oxidising & reducing powers : The metals placed above hydrogen in the electrochemical series are strong reducing agents whereas non-metals placed after hydrogen, are strong oxidising agents.

(v) Displacement of one non-metal from its salt solution by another non-metal : A non-metal lower in the series will have more reduction potential and will displace another non-metal with lower reduction potential. e.g. F_2 can displace all halide ion from solution.

$F_2 + 2KCl \longrightarrow 2KF + Cl_2$

$Cl_2 + 2KI \longrightarrow 2KCl + I_2$

Nernst Equation :

For a reaction $aA + bB \rightleftharpoons cC + dD$

$\Delta G = \Delta G^\circ + RT\ln Q$

$-nFE = -nFE^\circ + RT\ln Q$

With the help of Nernst equation, we can calculate the non-standard electrode potential of electrode or EMF of cell. Nernst equation predicts effects of concentration, pressure or temperature changes on cell EMF. Nernst equation can be applied on half-cell as well as complete Galvanic cell reaction.

$$E_{cell} = E^0 - \frac{RT}{nF}\ln Q = E^0 - \frac{2.303RT}{nF}\log Q$$

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Where - E^0 = Standard electrode potential

R = Gas constant

T = Temperature (in K)

F = Faraday (96500 coulomb mol^{-1})

n = No. of $e^\ominus$ gained or lost in balanced equation.

Q = Reaction quotient

$\frac{2.303\times 8.314\times 298}{96500}$ = 0.059 volt (At 298 K)

Note:

(i) For writing Nernst equation, first write balanced cell reaction.

(ii) Nernst equation can be applied on half-cell as well as complete Galvanic cell reaction.

Thermodynamic treatment of cell :

(i) Determination of equilibrium constant : We know, that

$$E = E^0 - \frac{0.0591}{n}\log Q \qquad \text{... (i)}$$

At equilibrium, the cell potential is zero because cell reactions are at equilibrium, i.e. E = 0 & Q = K_{eq}.

$\therefore$ From (i), we have

$$0 = E^0 - \frac{0.0591}{n}\log K_{eq} \text{ or } \qquad K_{eq} = \text{anti log}\left[\frac{nE^0}{0.0591}\right] = 10^{\frac{nE^0}{0.0591}}$$

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(ii) Heat of Reaction inside the cell: Let n Faraday charge flows out of a cell of e.m.f. E, then

$-\Delta G = nFE$ ... (i)

Gibbs Helmholtz equation (from thermodynamics) may be given as,

$$\Delta G = \Delta H + T\left[\frac{\partial \Delta G}{\partial T}\right]_p \quad \text{... (ii)}$$

From Eqs. (i) and (ii), we have

$$-nFE = \Delta H + T\left[\frac{\partial(-nFE)}{\partial T}\right]_p = \Delta H - nFT\left[\frac{\partial E}{\partial T}\right]_p$$

$$\therefore \quad \Delta H = -nFE + nFT\left[\frac{\partial E}{\partial T}\right]_p$$

(iii) Entropy change inside the cell : We know that G = H - TS or $\Delta G = \Delta H - T\Delta S$ ...(i)

where ΔG = Free energy change ; ΔH = Enthalpy change and ΔS = entropy change.

According to Gibbs Helmoholtz equation,

$$\Delta G = \Delta H + T\left[\frac{\partial \Delta G}{\partial T}\right]_p \quad \text{...(ii)}$$

From Eqs. (i) and (ii), we have

$$-T\Delta S = T\left[\frac{\partial \Delta G}{\partial T}\right]_p \quad \text{or} \quad \Delta S = -\left[\frac{\partial \Delta G}{\partial T}\right]_p \quad \text{or} \quad \Delta S = nF\left[\frac{\partial E}{\partial T}\right]_p$$

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where $\left[\frac{\partial E}{\partial T}\right]_p$ is called temperature coefficient of cell.

Electrode concentration cell :

$Pt(s) \mid H_2(P_1) \mid H^{\oplus}(C) \mid H_2(P_2) \mid Pt(s)$

Here, hydrogen gas is bubbled at two different partial pressures at electrode dipped in the solution of same electrolyte.

Anode : $\frac{1}{2}H_2(p_1) \rightarrow H^{\oplus}(C) + e^{\ominus}$

Cathode : $H^{\oplus}(C) + e^{\ominus} \rightarrow \frac{1}{2}H_2(p_2)$

Final cell reaction : $\frac{1}{2}H_2(p_1) \rightleftharpoons \frac{1}{2}H_2(p_2)$

$$\therefore \quad E = -\frac{2.303RT}{1\times F}\log\left[\frac{p_2}{p_1}\right]^{1/2}$$

$$\text{or} \quad E = -\left[\frac{2.303\,RT}{2F}\right]\log\left[\frac{p_2}{p_1}\right],$$

$$\text{At } 25^\circ C, \; E = \frac{0.059}{2F}\log\left[\frac{p_1}{p_2}\right]$$

For spontaneity of such cell reaction, $p_1 > p_2$

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Electrolyte concentration cells :

$Zn(s) \mid ZnSO_4\,(C_1) \parallel ZnSO_4\,(C_2) \mid Zn(s)$

In such cells, concentration gradient rise in electrolyte solutions. Cell process may be given as,

$Zn(s) \rightarrow Zn^{2\oplus}(C_1) + 2e$ (Anodic reaction)

$Zn^{2\oplus}(C_2) + 2e \longrightarrow Zn(s)$ (Cathodic reaction)

$Zn^{2\oplus}(C_2) \rightleftharpoons Zn^{2\oplus}(C_1)$ (Final cell reaction)

$\therefore$ From Nernst equation, we have

$$E = 0 - \frac{2.303RT}{2F}\log\left[\frac{C_1}{C_2}\right] \text{ or } E = \frac{2.303RT}{2F}\log\left[\frac{C_2}{C_1}\right]$$

For spontaneity of such cell reaction, $C_2 > C_1$

Illustration 1:

E° of some oxidants are given as :

$I_2 + 2e^{\ominus} \longrightarrow 2I^{\ominus}$, E° = + 0.54 V

$MnO_4^{\ominus} + 8H^{\oplus} + 5e^{\ominus} \longrightarrow Mn^{2\oplus} + 4H_2O$, E° = + 1.52 V

$Fe^{3\oplus} + e^{\ominus} \longrightarrow Fe^{2\oplus}$, E° = + 0.77 V

$Sn^{4\oplus} + 2e^{\ominus} \longrightarrow Sn^{2\oplus}$, E° = + 0.1 V

(a) Select the strongest reductant and oxidant in these.

(b) Select the weakest reductant and oxidant in these.

(c) Select the spontaneous reaction from the changes given below.

(i) $Sn^{4\oplus} + 2Fe^{2\oplus} \longrightarrow Sn^{2\oplus} + 2Fe^{3\oplus}$

(ii) $2Fe^{2\oplus} + I_2 \longrightarrow 2Fe^{3\oplus} + 2I^{\ominus}$

(iii) $Sn^{4\oplus} + 2I^{\ominus} \longrightarrow Sn^{2\oplus} + I_2$

(iv) $Sn^{2\oplus} + I_2 \longrightarrow Sn^{4\oplus} + 2I^{\ominus}$

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Solution:

(a) More the E^{o}_{OP}, more is the tendency for oxidation. Therefore, since maximum E^{o}_{OP} stands for :

$Sn^{2\oplus} \longrightarrow Sn^{4\oplus} + 2e^{\ominus}$ $E^{o}_{OP} = -0.1V$

$\therefore$ Strongest reductant : $Sn^{2\oplus}$

and weakest oxidant : $Sn^{4\oplus}$

(b) More positive is E^{o}_{RP}, more is the tendency for reduction. Therefore, since maximum E^{o}_{RP} stands for :

$MnO_4^{\ominus} + 8H^{\oplus} + 5e^{\ominus} \longrightarrow Mn^{2\oplus} + 4H_2O$ $E^{o}_{RP} = +1.52$ V

$\therefore$ Strongest oxidant : $MnO_4^{\ominus}$

and weakest reductant: $Mn^{2\oplus}$

Note : Stronger is oxidant, weaker is its conjugate reductant and vice-versa.

(c) For (i) $E^{o}_{Cell} = E^{o}_{OP_{Fe^{2\oplus}/Fe^{3\oplus}}} + E^{o}_{RP_{Sn^{4\oplus}/Sn^{2\oplus}}} = -0.77 + 0.1$

$Fe^{2\oplus}$ oxidizes and $Sn^{4\oplus}$ reduces in change.

$\therefore$ $E^{o}_{Cell} = E^{o}_{OP_{Fe^{2\oplus}/Fe^{3\oplus}}} + E^{o}_{RP_{Sn^{4\oplus}/Sn^{2\oplus}}} = -0.77 + 0.1 = -0.67$ V

E^{o}_{Cell} is negative.

$\therefore$ (i) Is non-spontaneous change.

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For (ii) $E^{o}_{Cell} = E^{o}_{OP_{Fe^{2\oplus}/Fe^{3\oplus}}} + E^{o}_{RP_{I_2/I^{\ominus}}} = -0.77 + 0.54 = -0.23\ V$

$\therefore$ (ii) reaction is non-spontaneous change.

For (iii) $E^{o}_{Cell} = E^{o}_{OP_{I^{\ominus}/I_2}} + E^{o}_{RP_{Sn^{4\oplus}/Sn^{2\oplus}}} = -0.54 + 0.1 = -0.44\ V$

$\therefore$ (iii) Reaction is non-spontaneous change.

For (iv) $E^{o}_{Cell} = E^{o}_{OP_{Sn^{2\oplus}/Sn^{4\oplus}}} + E^{o}_{RP_{I_2/I}} = -0.1 + 0.54 = +0.44\ V$

(iv) Reaction is spontaneous change.

Illustration 2:

Given the standard electrode potentials ;

$K^{\oplus}|K = -2.93$ V, $Ag^{\oplus}|Ag = 0.80$ V, $Hg^{2\oplus}|Hg = 0.79$ V, $Mg^{2\oplus}|Mg = -2.37$ V, $Cr^{3\oplus}|Cr = -0.74$V.

Arrange these metals in their increasing order of reducing power.

Solution:

More is $E°_{RP}$, more is the tendency to get reduced or more is the oxidising power or lesser is reducing power. Ag < Hg < Cr < Mg < K

Illustration 3:

A cell is prepared by dipping a copper rod in 1 M $CuSO_4$ solution and a nickel rod in 1M $NiSO_4$. The standard reduction potentials of copper and nickel electrodes are + 0.34 V and –0.25 V respectively.

(i) Which electrode will work as anode and which as cathode ?

(ii) What will be the cell reaction ?

(iii) How is the cell represented ?

(iv) Calculate the EMF of the cell.

Solution:

(i) The nickel electrode with smaller E° value (–0.25 V) will work as anode while copper electrode with more E° value (+0.34V) will work as cathode.

(ii) The cell reaction may be written as :

At anode : $Ni\ (s) \longrightarrow Ni^{2\oplus}\ (aq.) + 2e^{\ominus}$

At cathode : $Cu^{2\oplus}\ (aq) + 2e^{\ominus} \longrightarrow Cu\ (s)$

Cell reaction : $Ni\ (s) + Cu^{2\oplus}\ (aq) \longrightarrow Ni^{2\oplus}\ (aq) + Cu\ (s)$

(iii) The cell may be represented as :

$Ni(s)|Ni^{2\oplus}\ (aq)\ ||\ Cu^{2\oplus}\ (aq)|Cu\ (s)$

(iv) EMF of cell = $E^{o}_{cathode} - E^{o}_{anode} = (+0.34) - (-0.25) = 0.59$ V

Illustration 4:

Predict whether the following reaction can occur under standard conditions or not.

$Sn^{2\oplus}\ (aq) + Br_2\ (\ell) \longrightarrow Sn^{4\oplus}\ (aq) + 2Br^{\ominus}(aq)$

Given : $E^{o}_{Sn^{4\oplus}|Sn^{2\oplus}} = +0.15$; $E^{o}_{Br_2|Br^{\ominus}} = 1.06$V.

Solution:

$E^{o}_{cell} = E^{o}_{cathode} - E^{o}_{anode} = 1.06 - 0.15 = 0.91$V.

Since, E^{o}_{cell} comes out be positive, this means that the reaction can occur.

Illustration 5:

Given that, $Co^{3\oplus} + e^{\ominus} \longrightarrow Co^{2\oplus}$; $E° = +1.82$V

$2H_2O \longrightarrow O_2 + 4H^{\oplus} + 4e^{\ominus}$; $E° = -1.23$V.

Explain why $Co^{3\oplus}$ is not stable in aqueous solutions.

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Solution:

The E^{o}_{cell} can be calculated as follows :

$4 \times [Co^{3\oplus} + e^{\ominus} \longrightarrow Co^{2\oplus}]$; $E^{\circ} = +1.82V$

$1 \times [2H_2O \longrightarrow O_2 + 4H^{\oplus} + 4e^{\ominus}]$; $E^{\circ} = -1.23\ V.$

Add : $4Co^{3\oplus} + 2H_2O \longrightarrow 4Co^{2\oplus} + 4H^{\oplus} + O_2$; $E^{\circ} = 1.82 - 1.23 = +0.59V.$

Since, E^{o}_{cell} is positive, the cell reaction is spontaneous. This means that $Co^{3\oplus}$ ions will take part in the reaction. Therefore, $Co^{3\oplus}$ is not stable.

Illustration 6:

The 0.1 M copper sulphate solution in which copper electrode is dipped at 25°C. Calculate the electrode reduction potential of copper electrode.

[Given : $E^{o}_{Cu^{2\oplus}|Cu} = 0.34V$]

Solution:

$Cu^{2\oplus} + 2e^{\ominus} \rightarrow Cu$

$$E_{red} = E^{o}_{red} - \frac{0.059}{n}\log Q = E^{o}_{Cu^{2\oplus}|Cu} - \frac{0.0591}{2}\log\frac{1}{[Cu^{2\oplus}]}$$

So, $E = 0.34 - \frac{0.0591}{2}\log 10$

$= 0.34 - 0.03 = 0.31$ volts

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Illustration 7:

Calculate the EMF of the cell : $Cr|Cr^{3\oplus}$ (0.1M) || $Fe^{2\oplus}$ (0.01M)|Fe

(Given : $E^{o}_{Cr^{3\oplus}|Cr} = -0.75$ V, $E^{o}_{Fe^{2\oplus}|Fe} = -0.45$ V)

Solution:

Half cell reactions are :

- **At anode :** $[Cr \rightarrow Cr^{3\oplus} + 3e^{\ominus}] \times 2$
- **At cathode :** $[Fe^{2\oplus} + 2e^{\ominus} \rightarrow Fe] \times 3$

Over all reaction : $2Cr(s) + 3Fe^{2\oplus}(aq.) \rightarrow 2Cr^{3\oplus}(aq.) + 3Fe(s.)$

$E^{o} = E^{o}_{c} - E^{o}_{A} = -0.45 - (-0.75) = 0.30$ V

$$E_{cell} = E^{0} - \frac{0.0591}{n}\log Q = 0.30 - \frac{0.0591}{6}\log\frac{[Cr^{3\oplus}]^2}{[Fe^{2\oplus}]^3} = 0.30 - \frac{0.0591}{6}\log\frac{[0.1]^2}{[0.01]^3} = 0.26\ V$$

Illustration 8:

The measured e.m.f. at 25°C for the cell reaction,

$Zn(s) + Cu^{2\oplus}_{(aq)}(1.0M) \longrightarrow Cu(s) + Zn^{2\oplus}_{(aq)}(0.1\ M)$ is 1.3 volt, Calculate E° for the cell reaction.

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Solution:

Using Nernst equation (at 298 K), $E_{cell} = E^{o}_{cell} - \frac{0.059}{2}\log\frac{[Zn^{2\oplus}(aq)]}{[Cu^{2\oplus}(aq)]}$

Here, E_{cell} = 1.3 V, $[Cu^{2\oplus}(aq)]$ = 1.0 M, $[Zn^{2\oplus}(aq)]$ = 0.1 M, = ?

Substituting the values, E^{o}_{cell} = 1.27 V

Illustration 9:

The emf of a cell corresponding to the reaction

$Zn + 2H^{\oplus}(aq) \longrightarrow Zn^{2\oplus}(0.1M) + H_2(g)$ 1 atm

is 0.28 volt at 25°C. Calculate the pH of the solution at the hydrogen electrode.

$E^{o}_{Zn^{2\oplus}|\,Zn}$ = – 0.76 volt and $E^{\circ}_{H^{\oplus}|H_2} = 0$

Solution:

$E^{\circ}_{cell} = 0.76\,volt$

Applying Nernst equation, $E_{cell} = E^{o}_{cell} - \frac{0.0591}{2}\log\frac{[Zn^{2\oplus}]P_{H_2}}{[H^{\oplus}]^2}$

$0.28 = 0.76 - \frac{0.0591}{2}\log\frac{(0.1)\times 1}{[H^{+}]^2}$, pH = 8.62

Illustration 10:

The half-cell oxidation potential of a half-cell $A^{x\oplus}$, $A^{(x+n)\oplus}$ | Pt were found to be as follows :

% of reduced form	24.4	48.8
Cell potential |V|	0.101	0.115

Determine the value of n. [Take $\frac{2.303RT}{F} = 0.06$, $\log_{10}$ 24.4 = 1.387, $\log_{10}$ 75.6 = 1.878, $\log_{10}$ 48.8 = 1.688, $\log_{10}$ 51.2 = 1.709]

Solution:

The half-cell reaction is -

$A^{x\oplus} \rightarrow A^{(x+n)\oplus} + ne^{\ominus}$

Its Nernst equation is -

$E = E^{\circ} - 2.303\,\frac{RT}{nF}\log\frac{[A^{(x+n)\oplus}]}{[A^{x\oplus}]} = E^{\circ} - \left(\frac{0.06V}{n}\right)\log\left(\frac{\text{oxidized form}}{\text{reduced form}}\right)$

Substituting the given values, we get

$0.101\ V = E^{\circ} - \left(\frac{0.06V}{n}\right)\log\frac{75.6}{24.4} = E^{\circ} - \left(\frac{0.06\,V}{n}\right)(0.491)$ ... (i)

$0.115\ V = E^{\circ} - \left(\frac{0.06V}{n}\right)\log\frac{51.2}{48.8} = E^{\circ} - \left(\frac{0.06\,V}{n}\right)(0.021)$ ... (ii)

eq. (ii) – (i), n = 2

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Illustration 11:

What is the standard electrode potential for the electrode $MnO_4^{\ominus}$ | MnO_2 in solution $E^{\circ}_{MnO_4^{\ominus}|Mn^{2\oplus}}$ = 1.51 volt,

$E^{\circ}_{MnO_2|Mn^{2\oplus}}$ = 1.23 volt

Solution:

$MnO_4^{\ominus} + 8H^{\oplus} + 5e^{\ominus} \longrightarrow Mn^{2\oplus} + 4H_2O$; $E^{\circ} = 1.51$ volt

$\underline{MnO_2 + 4H^{\oplus} + 2e^{\ominus} \longrightarrow Mn^{2\oplus} + 2H_2O}$; $E^{\circ} = 1.23$ volt

$MnO_4^{\ominus} + 4H^{\oplus} + 3e^{\ominus} \longrightarrow MnO_2 + 2H_2O$; $E^{\circ} = ?$

$$E^{\circ} = \frac{5\times1.51 - 2\times1.23}{3} = \frac{7.55 - 2.45}{3} = \frac{5.09}{3} = 1.70 \text{ volt}$$

Illustration 12:

Calculate ΔG° for the reaction : $Cu^{2\oplus}$ (aq) + Fe (s) $\rightleftharpoons$ $Fe^{2\oplus}$(aq) + Cu (s).

Given that $E^{o}_{Cu^{2\oplus}|Cu}$ = + 0.34 V, $E^{o}_{Fe^{\oplus 2}/Fe}$ = – 0.44 V

Solution:

The cell reactions are :

$Fe\ (s) \longrightarrow Fe^{2\oplus}\ (aq) + 2e^{\ominus}$ ... at (anode)

$Cu^{2\oplus}\ (aq) + 2e^{\ominus} \longrightarrow Cu\ (s)$ ... at (cathode)

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We know that : $\Delta G^{\circ} = -nF\ E^{o}_{cell}$; n = 2

$$E^{o}_{cell} = \left[E^{o}_{(Cu^{2\oplus}/Cu)} - E^{o}_{(Fe^{2\oplus}/Fe)}\right] = (+0.34\text{ V}) - (-0.44\text{ V}) = +0.78\text{ V and F} = 96500\text{ C}$$

$\therefore\quad \Delta G^{\circ} = -nF\ E^{o}_{cell} = -(2) \times (96500\text{ C}) \times (+0.78\text{ V}) = -150540\text{ CV} = -150540\text{ J}$

Illustration 13:

At 298 K the standard free energy of formation of $H_2O(l)$ is– 256.5 kJ|mol & $OH^{\ominus}$ is 80 kJ|mol. What will be emf at 298 K of the cell Pt(s) | H_2(g, 1 bar) | $H^{\oplus}$(1M) | | $OH^{\ominus}$(1M) | O_2 (g, 1 bar) | Pt(s)

Solution:

Anode : $H_2 \longrightarrow 2H^{\oplus} + 2e^{\ominus}$

Cathode : $\frac{1}{2}O_2 + H_2O + 2e^{\ominus} \longrightarrow 2OH^{\ominus}$

Over all reaction : $H_2 + H_2O + \frac{1}{2}O_2 \longrightarrow 2H^{\oplus} + 2OH^{\ominus}$

$\Delta G^{\circ} = -256.5 + 2 \times 80 = -96.5$ kJ

$-\Delta G^{\circ} = nFE^{\circ}$

$+ 96.5 \times 1000 = 2 \times 96500 \times E^{\circ}$

$E^{\circ} = 0.5$ Volt

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Illustration 14:

Calculate the equilibrium constant for the reaction at 298 K.

$Zn\,(s) + Cu^{2\oplus}\,(aq) \rightleftharpoons Zn^{2\oplus}(aq) + Cu\,(s)$

Given, $E^{o}_{Zn^{2\oplus}|Zn} = -0.76\ V$ and $E^{o}_{Cu^{2\oplus}|Cu} = +0.34\ V$

Solution:

We know that, $\log K_c = \dfrac{nE^{o}_{cell}}{0.0591}$

$E^{o}_{cell} = \left[E^{o}_{Cathode} - E^{o}_{Anode}\right]$ = [(+0.34 V) – (– 0.76 V)] = 1.10 V, n = 2,

$\therefore \quad \log K_C = \dfrac{2\times(1.10\ V)}{(0.0591\ V)} = 37.29$, K_c = Antilog 37.29 = 1.95×10^{37}

Illustration 15:

Calculate the cell e.m.f. and ΔG for the cell reaction at 298 K for the cell.

$Zn\,(s)\,|\,Zn^{2\oplus}\,(0.0004M)\,||\,Cd^{2\oplus}\,(0.2M)\,|\,Cd(s)$

Given, $E^{o}_{Zn^{2\oplus}|Zn}$ = – 0.763 V ; $E^{o}_{Cd^{2\oplus}|Cd}$ = – 0.403 V at 298 K. F = 96500 C mol^{-1}.

Solution:

Step I. Calculation of cell e.m.f. :

According to Nernst equation,

$$E = E^{\circ} - \frac{0.0591}{n}\log\frac{[Zn^{2\oplus}(aq)]}{[Cd^{2\oplus}(aq)]}$$

$E^{o}_{cell} = E^{o}_{(Cd^{2\oplus}/Cd)} - E^{o}_{(Zn^{2\oplus}/Zn)}$ = (– 0.403) – (– 0.763) = 0.36 V

$[Zn^{\oplus}(aq)]$ = 0.0004 M, $[Cd^{2\oplus}\,(aq)]$ = 0.2 M, n = 2

$$E = (0.36) - \frac{(0.059V)}{2}\log\frac{0.0004}{0.2} = 0.36 - \frac{(0.059\ V)}{2}\times(-2.69990) = 0.36\ V + 0.08 = 0.44\ V$$

Step II. Calculation of ΔG :

$\Delta G = -\,nFE_{cell}$

E_{cell} = 0.44 V, n = 2 mol, F = 96500 c mol^{-1}

$\therefore \quad \Delta G = -$ (2 mol) × (96500 C mol^{-1}) × (0.44 V)

= – 84920 CV = – 84920 J

Illustration 16:

The cell Pt | H_2(g) (1 atm) | $H^{\oplus}$; pH = x | | Normal calomal electrode has EMF of 0.64 volt at 25°C. The standard reduction potential of normal calomal electrode is 0.28 V. What is the pH of solution in anodic compartment. Take $\dfrac{2.303RT}{F} = 0.06$ at 298 K.

Solution:

$$E_{H^{\oplus}/H_2} = -0.06\log\frac{1}{[H^{\oplus}]} = -0.06pH$$

$\Rightarrow 0.64 = E_{cathode} - E_{Anode} = 0.28 - (-\,0.06\ pH)$

$$\Rightarrow pH = \frac{0.64-0.28}{0.06} = \frac{0.36}{0.06} = 6$$

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Illustration 17:

Consider a Galvenic cell,

$Zn(s) \mid Zn^{2\oplus}(0.1\ M) \mid\mid Cu^{2\oplus}(0.1\ M) \mid Cu(s)$

by what factor, the electrolyte in anodic half-cell should be diluted to increase the emf by 9 milli volt at 298 K.

Solution:

$$E = E^\circ_{cell} - \frac{0.06}{2}\log\frac{[Zn^{2\oplus}]}{[Cu^{2\oplus}]}$$

$$E_1 = E^\circ_{cell} - 0.03\ \log\frac{(0.1)}{(0.1)}$$

$$E_2 = E^\circ_{cell} - 0.03\ \log\frac{(0.1/x)}{(0.1)}$$ {x is the factor by which electrolyte is diluted.}

$$E_2 - E_1 = 9 \times 10^{-3} = 0 - 0.03 \log\left(\frac{1}{x}\right)$$

$0.009 = 0.03 \log X$

$$\frac{9\times10^{-3}}{3\times10^{-2}} = 0.3 = \log X$$

$X = 2$

Illustration 18:

A disposable galvanic cell $Zn\left|Zn^{2\oplus}\right\|Sn^{2\oplus}\left|Sn\right.$ is produced using 1.0 mL of 0.5 M $Zn(NO_3)_2$ and 1.0 mL of 0.50 M $Sn(NO_3)_2$. It is needed to power a pace-maker that draws a constant current of 10^{-6} Amp to run it and requires atleast 0.50 V to function. Calculate the value of $[Zn^{2\oplus}]$ when cell reaches 0.5 V at 298 K.

(Given : $E^\circ\ (Zn^{2\oplus}|Zn) = -0.76V$; $E^\circ\ (Sn^{2\oplus}|Sn) = -0.14$ V).

Solution:

$$E = 0.5\ V = 0.62 - \frac{0.059}{2}\log\frac{0.5+x}{0.5-x}$$

$$\because \quad Zn + \underset{0.5-x}{Sn^{2\oplus}} \rightleftharpoons Sn + \underset{0.5+x}{Zn^{2\oplus}}$$

$$\Rightarrow \log\frac{0.5+x}{0.5-x} = \frac{0.12\times2}{0.059} \approx 4$$

$$\Rightarrow \frac{0.5+x}{0.5-x} = 10^4$$

or $x \approx 0.5$ M

$\therefore\ [Zn^{2\oplus}] = 1.0$ M

Illustration 19:

An alloy of Pb-Ag weighing 54 mg was dissolved in desired amount of HNO_3 & volume was made upto 500ml. An Ag electrode was dipped in solution and then connected to standard hydrogen electrode anode. Then calculate % of Ag in alloy.

Given : $E_{cell} = 0.5$ V ; $E^o_{Ag^{\oplus}|Ag} = 0.8V$, $\frac{2.303RT}{F} = 0.06$

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Solution:

$Ag^{\oplus} + 1\ e^{\ominus} \longrightarrow Ag$

$E_{cell} = 0.5 = 0.8 + \frac{0.06}{1}\log [Ag^{\oplus}]$

$\log [Ag^{\oplus}] = \frac{-0.30}{0.06} = -5$

$[Ag^{\oplus}] = 10^{-5}$ mol|L

moles of $Ag^{\oplus}$ in 500 ml = $\frac{10^{-5}}{2}$

Mass of Ag = $\frac{10^{-5}}{2} \times 108$

% Ag = $\frac{\frac{10^{-5}}{2} \times 108}{54 \times 10^{-3}} \times 100 = 1$

Illustration 20:

A solution contains $A^{\oplus}$ and $B^{\oplus}$ in such a concentration that both deposits simultaneously. If current of 9.65 amp was passed through 100 ml solution for 55 seconds then find the final concentration of $A^{\oplus}$ ions if initial concentration of $B^{\oplus}$ is 0.1M.

Given : $A^{\oplus} + e^{\ominus} \longrightarrow A$; $E° = -0.5$ volt

$B^{\oplus} + e^{\ominus} \longrightarrow B$; $E° = -0.56$ volt

$\frac{2.303RT}{F} = 0.06$

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Solution:

$-0.5 - \frac{0.06}{1}\log\frac{1}{[A^{\oplus}]} = -0.56 - \frac{0.06}{1}\log\frac{1}{[B^{\oplus}]}$

$0.06 = \frac{0.06}{1}\log\frac{[B^{\oplus}]}{[A^{\oplus}]}$

$\frac{[B^{\oplus}]}{[A^{\oplus}]} = 10$

$[A^{\oplus}]_{initial} = 0.01M$

Illustration 21:

While the discharging of a lead storage battery following reaction take place.

$PbO_2 + Pb + 4H^{\oplus} + 2SO_4^{2\ominus} \rightarrow 2PbSO_4 + 2H_2O$; $E° = 2.01$

Calculate the energy (in kJ) obtained from a lead storage battery in which 0.014 mol of lead is consumed. Assume a constant concentration of 10.M H_2SO_4 (log2 = 0.3)

Solution:

$E = E° - \frac{0.059}{n}\log\frac{1}{[H^{\oplus}]^4[SO_4^{2\ominus}]^2} = 2.01 - \frac{0.059}{2}\log\frac{1}{[20]^4[10]^2} = 2.22$ V

Energy = qE

= 2 × 0.014 × 96500 × 2.22

= 6000 J = 6 kJ

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Illustration 22:

Consider the following standard reduction potentials :-

$Fe^{2\oplus} + 2e^{\ominus} \rightleftharpoons Fe$; $E° = -0.41$ V

$Ag^{\oplus} + e^{\ominus} \rightleftharpoons Ag$; $E° = 0.80$ V

$O_2 + 2H_2O + 4e^{\ominus} \longrightarrow 4OH^{\ominus}$; $E° = 0.40$ V

What would happen if a block of silver metal is connected to a buried iron pipe via a wire:-

(A) The silver metal would corrode, a current would be produced in the wire, and O_2 would be reduced on the surface of the iron pipe.

(B) The silver metal would corrode, a current would be produced in the wire, and $Fe^{2\oplus}$ would be reduced on the surface of the iron pipe.

(C) The iron pipe would corrode, a current would be produced in the wire, and $Ag^{\oplus}$ would be reduced on the surface of the silver metal.

(D) The iron pipe would corrode, no current would be produced in the wire, and O_2 would be reduced on the surface of the iron pipe.

Ans. (D)

Solution:

The iron pipe would corrode, no current would be produced in the wire, and O_2 would be reduced on the surface of the iron pipe.

Electrolytic conductance:

Resistance (R) :

Metallic and electrolytic conductors obey ohm's law according to which the resistance of a conductor is the ratio of the applied potential difference (V) to the current flowing(I).

$$R = \frac{V}{I}$$

- R is expressed in ohms.
- In the case of solution of electrolytes, the resistance offered by the solution to the flow of current is –
  (a) Directly proportional to the distance between the electrodes
  $R \propto \ell$
  (b) Inversely proportional to the area of cross section of the electrodes
  $R \propto \frac{1}{A}$

Conductance (G) :

The conductance of a conductor is equal to reciprocal of resistance.

$G = \frac{1}{R}$ • G is expressed in mho or Ω^{-1} or Siemen(S). [$1S = 1\Omega^{-1}$ S.I. unit]

Specific resistance or resistivity (ρ) :

The resistance (R) of a conductor of uniform cross section is directly proportional to its length(ℓ) and inversely proportional to its area of cross section (A).

$$R \propto \frac{\ell}{A} \qquad R = \rho \frac{\ell}{A}$$

where ρ is a constant and called resistivity or specific resistance.

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When $\ell = 1$, A = 1 , then $\rho = R$ thus the specific resistance may be defined as the resistance of a conductor of unit length and unit area of cross section.

- Unit of $\rho \rightarrow$ ohm.cm

Specific conductance or conductivity (κ) :

It is defined as the reciprocal of specific resistance

$$\kappa = \frac{1}{\rho},$$

$$G = \kappa / G^* \ , G^* = \frac{l}{a} = \text{cell constant}$$

If $\ell = 1$ cm & A = 1 cm^2 then $\boldsymbol{\kappa} = G$

Hence conductivity or specific conductance (κ) of a solution is defined as the conductance of one centimeter cube volume of the solution of the electrolyte.

- Cell constant is a fixed quantity for a particular cell and is defined as the distance between two parallel electrodes of a cell divided by the area of cross section of the electrodes.

  κ = G × cell constant

- Unit of $\kappa \rightarrow$ ohm^{-1} cm^{-1}
- SI unit of $\kappa \rightarrow$ S m^{-1}

  100 S m^{-1} = 1 ohm^{-1} cm^{-1}

Molar conductance (λm or Λm) :

- It is defined as the product of specific conductance (κ) and the volume (V in mL) which contains one mole of the electrolyte.

  $\Lambda_m = \kappa \times v$ and $\Lambda_m = \frac{\kappa \times 1000}{M}$ **ohm^{-1} cm^2 mol^{-1}** **[SI unit : S m^2 mol^{-1}]**

- It can also be defined as conductance of 1 mole electrolyte completely dissolved between two plates separated by unit distance.

Equivalent conductance (λeq or Λeq) :

- It is defined as the product of specific conductance (κ) and the volume (V in mL) which contains one equivalent of the electrolyte.

  $\Lambda_{eq} = \kappa \times v$ and $\Lambda_{eq} = \frac{\kappa \times 1000}{N}$ ohm^{-1} cm^2 eq^{-1} [SI unit : S m^2 eq^{-1}]

- It can also be defined as conductance of 1 equivalent electrolyte completely dissolved between two plates separated by unit distance.

- **Relation between $\Lambda_{eq.}$ and Λ_m :**

  $$\Lambda_m = \frac{\kappa \times 1000}{M} \text{ and } \Lambda_{eq.} = \frac{\kappa \times 1000}{N}$$

  We know that, **Normality = n-factor × Molarity**

  or $N = n \times M \Rightarrow \boxed{\lambda_{eq} = \frac{\lambda_M}{n}}$

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n-factor = total cationic (or anionic) charge of salt.

Ex. $\Lambda_{eq}[Al_2(SO_4)_3] = \frac{\Lambda_m[Al_2(SO_4)_3]}{6}$, $\Lambda_{eq\ NaCl} = \frac{\Lambda_{m\ NaCl}}{1}$, $\Lambda_{eq.}CaCl_2 = \frac{\Lambda_{m\ CaCl_2}}{2}$

Effect of dilution on the conductivity of electrolytes :

(i) The degree of ionisation of weak electrolytes increases with the increase of dilution of the solution the conductivity is increased due to increase in the number of ions.

(ii) Effect of dilution on specific conductance :

Specific conductance decreases with the increase in dilution of the solution. On dilution, the no. of ions in 1 cm^3 solution decreases hence conductance also decreases.

(iii) Effect of dilution on equivalent or molar conductivity :

The equivalent or molar conductivity increases with dilution. For strong electrolyte λ_m or λ_{eq} increases very slowly but for weak electrolytes λ_m & λ_{eq} increase sharply on dilution.

- When the whole of the electrolyte has ionised, further addition of the water brings a small change in the value of equivalent or molar conductance. This stage is called **infinite dilution**.
- The ratio of equivalent conductivity at any dilution to equivalent conductivity at infinite dilution is called **conductivity ratio** or degree of dissociation of solute -

$$\alpha = \frac{\lambda_{eq.}}{\lambda_{eq}^{\infty}} = \frac{\lambda_m}{\lambda_m^{\infty}}$$

[λ_m^{∞} = molar conductance at ∞ dilution.]

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Variation of conductivity and molar conductivity with concentration :

- Conductance and conductivity always decrease with the decrease in concentration both for weak and strong electrolytes.
- It is because the number of ions per unit volume that carry the current in a solution decreases on dilution decreasing, I, C & κ.

Strong Electrolytes :

- For strong electrolytes, Λ increases slowly with dilution and can be represented by the equation

  $\Lambda = \Lambda^\circ - B\sqrt{C}$, B = constant (Debye-Huckel-Onsager equation)

  On dilution interionic separation increases causing free movement & less hindrance. This increase Λ_m & $\Lambda_{eq.}$

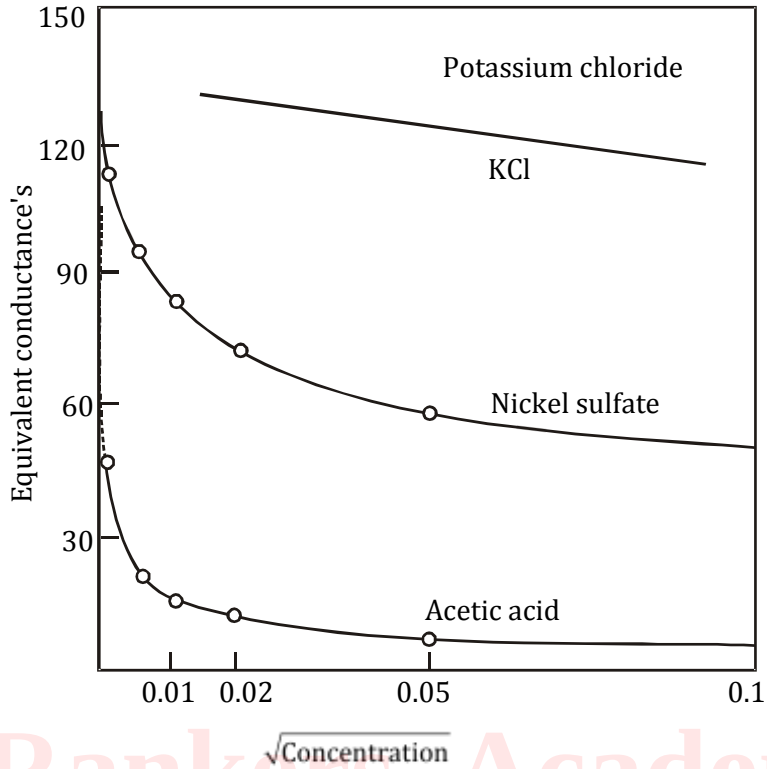
  For strong electrolytes Λ° or Λ^∞ can be calculated graphically from y-intercept.

  It can be seen that if we plot, Λ_m against $\sqrt{C}$, we obtain a straight line with intercept equal to Λ°_m and slope equal to B. The value of the constant B for a given solvent and temperature depends on the type of the electrolyte i.e., the charges on the cation and anion produced on the dissociation of the electrolyte in the solution. Thus, NaCl, $CaCl_2$, $MgSO_4$ are known as 1 – 1, 2 – 1 and 2 – 2 electrolytes respectively. All electrolytes of a particular type have the same value for B.

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Weak electrolytes :

- Weak electrolytes like acetic acid have lower degree of dissociation at higher concentration and hence for such electrolytes, the change in Λ with dilution is due to increase in the number of ions solution containing a given amount of electrolyte.



$KCl \rightarrow$ Strong
$Ni_2SO_4 \rightarrow$ Moderate
$CH_3COOH \rightarrow$ Weak electrolyte

Conductance of electrolytes of different types

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Note :

(A) Weaker the electrolyte more sharp will be increase of Λ_m or $\Lambda_{eq.}$ on dilution.

(B) Same plot is also observed for Λ_m vs. $\sqrt{\text{molarity}}$ of respective electrolytes.

Kohlrausch's law :

- "At infinite dilution, when dissociation is complete, each ion makes a definite contribution towards equivalent conductance of the electrolyte irrespective of the nature of the ion with which it is associated and the value of equivalent conductance at infinite dilution for any electrolyte is the sum of contribution of its constituent ions."

i.e., for $MgCl_2$; $\Lambda^{\infty}_{eq.} = \Lambda^{\infty}_{eq.}(Mg^{2\oplus}) + \Lambda^{\infty}_{eq.}(Cl^{\ominus})$

$$\Lambda^{\infty}_{m} = \Lambda^{\infty}_{m}(Mg^{2\oplus}) + \Lambda^{\infty}_{m}(Cl^{\ominus}) \times 2$$

$$\Lambda^{\infty}_{eq} = \frac{\Lambda^{\infty}_{m}(Mg^{2\oplus})}{2} + \frac{\Lambda^{\infty}_{m}(Cl^{\ominus})}{1}$$

Note :

$\lambda^{\infty} = \lambda^{\circ}$ [Both are used interchangeably]

$$\lambda^{0}_{eq}(\text{cation}) = \frac{\lambda^{0}_{m}}{\text{charge on the cation}}$$

ex. $\lambda^{0}_{eq}\,Al^{3\oplus} = \frac{\lambda^{0}_{m}Al^{3\oplus}}{3}$

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$$\lambda^0_{eq}(\text{anion}) = \frac{\lambda^0_m}{\text{charge on the anion}}$$

$$\Lambda^0_{eq}, \text{electrolyte} = \frac{\lambda^0_m \text{ electrolyte}}{\text{total +ve charge on cations in electrolyte}}$$

or

total –ve charge on anions in electrolyte

The Independent Migration of Ions. A survey of equivalent conductance's at infinite dilution of a number of electrolytes having an ion in common will bring to light certain regularities ;

Comparison of equivalent conductance's at infinite dilution :

Electrolyte	Λ_0	Electrolyte	Λ_0	Difference
KCl	130.0	NaCl	108.9	21.1
KNO_3	126.3	$NaNO_3$	105.2	21.1
K_2SO_4	133.0	Na_2SO_4	111.9	21.1

Observations of this kind were first made by **Kohlrausch (1879, 1885)** by comparing equivalent conductance at high dilutions; described them to the fact that under these conditions every ion makes a definite contribution towards the equivalent conductance of the electrolyte, irrespective of the nature of the other ion with which it is associated in the solution. The value of the equivalent conductance at infinite dilution may thus be regarded as made up of the sum of two independent factors, one characteristic of each ion; this result is known as Kohlrausch's law of independent migration of ions.

The ionic conductance is a definite constant for each ion, in a given solvent, its value depending only on the temperature.

It will be seen later that the ionic conductance at infinite dilution are related to the speeds with which the ions move under the influence of an applied potential gradient.

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Applications of Kohlrausch's law :

- **Calculate Λ° for any electrolyte from the Λ° of individual ions.**

An important use of ionic conductance is to determine the equivalent conductance at infinite dilution of certain electrolytes which cannot be, or have not been, evaluated from experimental data. For example, with a weak electrolyte the extrapolation to infinite dilution is very uncertain, and with sparingly soluble salts the number of measurements which can be made at appreciably different concentrations is very limited. The value of Λ° can, however, be obtained by adding the ionic conductance. For example, the equivalent conductance of acetic acid at infinite dilution is the sum of the conductance of the hydrogen and acetate ions; the former is derived from a study of strong acids and the latter from measurements on acetates. It follows, therefore, that at 25°C.

$$\Lambda^\circ_{(CH_3COOH)} = \lambda^0_{H^\oplus} + \lambda^0_{CH_3COO^\ominus} = 349.8 + 40.9 = 390.7 \text{ ohms}^{-1} \text{ cm}^2$$

The same result can be derived in another manner which is often convenient since it avoids the necessity of separating the conductance of an electrolyte into the contributions of its constituent ions. The equivalent conductance of any weak electrolyte MA at infinite dilution, therefore, can be written as :

$\Lambda^\circ(MA) = \Lambda^\circ(MCl) + \Lambda^\circ(NaA) - \Lambda^\circ(NaCl)$, [MCl, NaA, NaCl are strong electrolytes]

where Λ° (MCl), Λ°(NaA) and Λ°(NaCl) are the equivalent conductance at infinite dilution of the chloride of the metal M, i.e., MCI, of the sodium salt of the anion A, i.e., NaA, and of sodium chloride, respectively. Any convenient anion may be used instead of the chloride ion, and similarly the sodium

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ion may be replaced by other metallic cation or by the hydrogen ion. For example, if $M^{\oplus}$ is the hydrogen ion and $A^{\ominus}$ is the acetate ion, it follows that

$\Lambda°(CH_3COOH) = \Lambda°(HCl) + \Lambda°(CH_3COONa) - \Lambda°(NaCl)$

$= 426.16 + 91.0 - 126.45$

$= 390.71$ ohms^{-1} cm^2 at 25°C.

Similarly :

$\Lambda_m°[BaSO_4] = \Lambda_m°[BaCl_2] + \Lambda_m°[Na_2SO_4] - 2\Lambda_m°[NaCl]$

- **Degree of dissociation :**

$\therefore$ Degree of dissociation = $\dfrac{\text{equivalent conductance at a given concentration}}{\text{equivalent conductance at infinite dilution}}$

$$\alpha = \frac{\Lambda_m^c}{\Lambda_m^\infty} = \frac{\Lambda_{eq}^c}{\Lambda_{eq}^\infty}$$

- **Dissociation constant** of weak electrolyte :

$$K_C = \frac{C\alpha^2}{1-\alpha} \quad \frac{C\left(\frac{\Lambda_m^c}{\Lambda_m^\infty}\right)^2}{1-\frac{\Lambda_m^c}{\Lambda_m^\infty}}$$

- **Solubility(s) and K_{SP} of any sparingly soluble salt.**

Sparingly soluble salt has very less solubility

Solubility = molarity $\cong S \to 0$

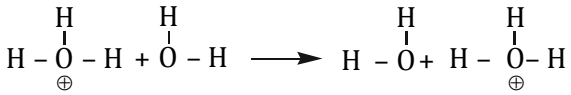
So, solution can be considered to be of zero conc. or infinite dilution.

$$\Lambda_m^\infty \simeq \Lambda_m^0 = \kappa \times \frac{1000}{s}$$

$$s = \frac{\kappa \times 1000}{\Lambda_m^0}$$

Abnormal ionic conductance of $H^{\oplus}$ and $OH^{\ominus}$:

It is supposed, as already indicated, that the hydrogen ion in water is $H_3O^{\oplus}$ with three hydrogen atoms attached to the central oxygen atom. When a potential gradient is applied to an aqueous solution containing hydrogen ions, the latter travel to some extent by the same mechanism as do other ions, but there is in addition another mechanism which permits of a more rapid ionic movement. This second process is believed to involve the transfer of a proton ($H^{\oplus}$) from a $H_3O^{\oplus}$ ion to an adjacent water molecule; thus



The resulting $H_3O^{\oplus}$ ion can now transfer a proton to another water molecule, and in this way the positive charge will be transferred a considerable distance in a short time. The electrical conductance will thus be much greater than that due solely to the normal mechanism.

Conductometric titration :

- The principle of conductometric titration is based on the fact that during the titration, one of the ions is replaced by the other and invariably these two ions differ in the ionic conductivity with the result that conductivity of the solution varies during the course of titration. The equivalence point may be

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located graphically by plotting the change in conductance as a function of the volume of titrant added. In order to reduce the influence of errors in the conductometric titration to a minimum, the angle between the two branches of the titration curve should be as small as possible (see Fig.). If the angle is very obtuse, a small error in the conductance data can cause a large deviation. The following approximate rules will be found.

- The smaller the conductivity of the ion which replaces the reacting ion, the more accurate will be the result. Thus, it is preferable to titrate a silver salt with lithium chloride rather than with HCl. Generally, cations should be titrated with lithium salts and anions with acetates as these ions have low conductivity.
- The larger the conductivity of the anion of the reagent which reacts with the cation to be determined, or vice versa, the more acute is the angle of titration curve.
- The titration of a slightly ionized salt does not give good results, since the conductivity increases continuously from the commencement. Hence, the salt present in the cell should be virtually completely dissociated; for a similar reason; the added reagent should also be as strong electrolyte. The main advantages to the conductometric titration are its applicability to very dilute, and coloured solutions and to system that involve relatively incomplete reactions. For example, neither a potentiometric, nor indicator method can be used for the neutralization titration of phenol ($K_a \times 10^{-10}$) a conductometric endpoint can be successfully applied.

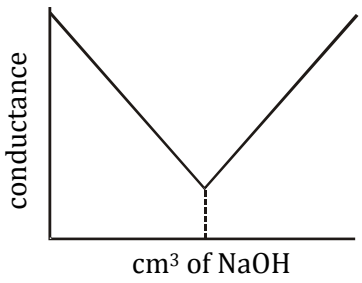
Cation	λ_m^∞ / (Ω^{-1} cm^2 mol^{-1})	Anion	λ_m^∞ / (Ω^{-1} cm^2 mol^{-1})
$H_3O^\oplus$	350.0	$OH^\ominus$	199.2
$NH_4^\oplus$	73.5	$Br^\ominus$	78.1
$K^\oplus$	73.5	$Cl^\ominus$	76.5
$Na^\oplus$	50.1	$NO^\ominus$	71.4
$Ag^\oplus$	62.1	$CHCOO^\ominus$	40.0
$Ca^{2\oplus}$	118.0	$SO_4^{2\ominus}$	159.6
$Mg^{2\oplus}$	106.1		

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Some Typical Conductometric Titration Curves are:

1. Strong acid with a strong base:

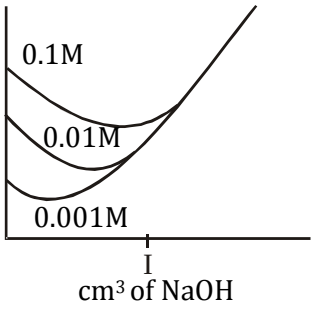
Before NaOH is added, the conductance is high due to the presence of highly mobile hydrogen ions. When the base is added, the conductance falls due to the replacement of hydrogen ions by the added cation as $H^\oplus$ ions react with $OH^\ominus$ ions to form undissociated water. This decrease in the conductance continues till the equivalence point. At the equivalence point, the solution contains only NaCl. After the equivalence point, the conductance increases due to the addition of NaOH.



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2. Weak Acid with a Strong Base, e.g. acetic acid with NaOH :

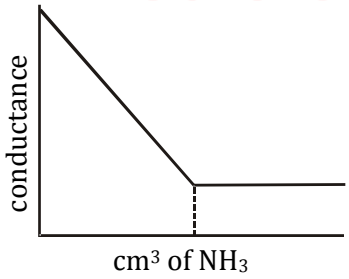
Initially the conductance is low due to the feeble ionization of acetic acid. On the addition of base, there is decrease in conductance not only due to the replacement of $H^{\oplus}$ by $Na^{\oplus}$ but also suppresses the dissociation of acetic acid due to common ion acetate. But very soon, the conductance increases on adding NaOH as NaOH neutralizes the un-dissociated CH_3COOH to CH_3OONa which is the strong electrolyte. This increase in conductance continues raise up to the equivalence point. The graph near the equivalence point is curved due the hydrolysis of salt CH_3COONa. Beyond the equivalence point, conductance increases more rapidly with the addition of NaOH due to the highly conducting $OH^{\ominus}$ ions.



3. Strong Acid with a Weak Base, e.g. sulphuric acid with dilute ammonia :

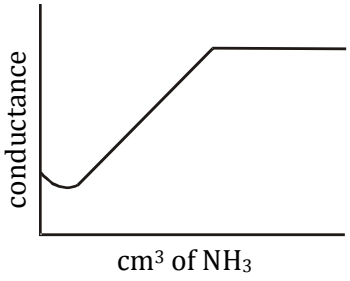
Initially the conductance is high and then it decreases due to the replacement of $H^{\oplus}$. But after the endpoint has been reached the graph becomes almost horizontal, since the excess aqueous ammonia is not appreciably ionised in the presence of ammonium sulphate.

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4. Weak Acid with a Weak Base :

The nature of curve before the equivalence point is similar to the curve obtained by titrating weak acid against strong base. After the equivalence point, conductance virtually remains same as the weak base which is being added is feebly ionized and, therefore, is not much conducting

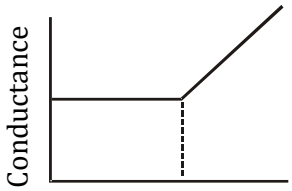


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5. Precipitation Titration and Complex Formation Titration :

Initially, the conductance was only due to the presence of $AgNO_3$ in the solution. When NaCl is added in it, $Ag^{\oplus}$ ions are getting replaced by $Na^{\oplus}$ ions and precipitate of AgCl is formed (which contributes negligibly in the conductance). Therefore, conductance doesn't change much.

After the equivalence point, conductance keeps on increasing because of further addition of NaCl.



$AgNO_3$(aq.) vs NaCl

Ionic mobility :

It is the speed of ion under unit potential gradient applied through solution.

$$= \frac{\text{speed of ion (s)}}{\text{Potential gradient}\left(\frac{dV}{dx}\right)} = \frac{cm^2}{\text{volt} - \text{sec.}} = \frac{\Lambda}{ZF}$$

$u/10^{-8}\ m^2s^{-1}v^{-1}$ in H_2O at 298 K			
$Li^{\oplus}$	4.01 4.65	$CHCOO^{\ominus}$	7.92
$Na^{\oplus}$	5.19	$F^{\ominus}$	5.74
$Cu^{2\oplus}$	5.47 5.56	$NO_3^{\ominus}$	7.40
$Ca^{2\oplus}$	6.17	$CO_3^{2\ominus}$	7.46
$Ag^{\oplus}$	6.24	$Cl^{\ominus}$	7.91
$NH_4^{\oplus}$	7.62	$I^{\ominus}$	7.96
$Rb^{\oplus}$	7.92	$SO_4^{2\ominus}$	8.29
$K^{\oplus}$	7.62	$OH^{\ominus}$	20.64
$H^{\oplus}$	36.23		

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Illustration 23:

Which of the following have maximum molar conductivity.

(i) 0.08 M solution and its specific conductivity is $2 \times 10^{-2}\ \Omega^{-1}\ cm^{-1}$.

(ii) 0.1 M solution and its resistivity is 50 Ω cm.

Solution:

(i) $\wedge_m = \frac{\kappa \times 1000}{M} = 2 \times 10^{-2} \times \frac{1000}{0.08} = 250\ \Omega^{-1}\ cm^2\ mol^{-1}$

(ii) $\wedge_m = \frac{\kappa \times 1000}{M}$ $\quad \because\ \kappa = \frac{1}{\rho}$ $\quad \therefore\ \Lambda_m \frac{1}{50} \times \frac{1000}{0.1} = 200\ \Omega^{-1}\ cm^2\ mol^{-1}$

So, the molar conductivity of 0.08 M solution will be greater than 0.1 M solution.

Illustration 24:

The equivalent conductivity of H_2SO_4 at infinite dilution is $384\ \Omega^{-1}\ cm^2\ eq^{-1}$. If 49 g H_2SO_4 per litre is present in solution and specific resistance is 18.4 Ω–cm then calculate the degree of dissociation.

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Solution:

Equivalent of $H_2SO_4 = \frac{49}{49} = 1$ N

Specific conductance $= \frac{1}{\text{specific resistance}} = \frac{1}{18.4}$

$\Rightarrow \lambda_{eq.} = \frac{1000 \times \kappa}{N} = \frac{1000}{18.4} = 55\ \Omega^{-1}\ cm^2\ eq^{-1}$

Degree of dissociation $(\alpha) = \frac{\lambda^C_{eq.}}{\lambda^{\infty}_{eq.}} = \frac{55}{384} = 0.14 \Rightarrow \alpha = 14\%$

Illustration 25:

Explain following ionic conductance data of 25°C for various fatty acid anions.

Anion	Formula	λ^0_- AT 25°C (in $ohm^{-1}cm^2eq^{-1}$)
Formate	$H CO_2^{\ominus}$	~52
Acetate	$CH_3 CO_2^{\ominus}$	40.9
Propionate	$CH_3CH_2 CO_2^{\ominus}$	35.8
Butyrate	$CH_3(CH_2)_2 CO_2^{\ominus}$	32.6
Valerianate	$CH_3(CH_2)_2 CO_2^{\ominus}$	~29
Caproate	$CH_3(CH_2)_4 CO_2^{\ominus}$	~28

Solution:

With increasing chain length bulk increases decreasing ionic mobility and thus equivalent conductance decreases.

$\because$ Charge is identical λ_m also decreases.

Illustration 26:

The resistance of a 0.01 N solution of an electrolyte was found to 210 ohm at 298 K using a conductivity cell with a cell constant of 0.88 cm^{-1}. Calculate specific conductance and equivalent conductance of solution.

Solution:

Given, for 0.01 N solution.

R = 210 ohm $\frac{\ell}{a}$, = 0.88 cm^{-1}

Specific conductance,

$\therefore \quad \kappa = \frac{1}{R} \times \frac{\ell}{a} \Rightarrow \kappa = \frac{1}{210} \times 0.88 = 4.19 \times 10^{-3}$ mho cm^{-1}

$\Lambda_{eq} = \frac{\kappa \times 1000}{N} = \frac{4.19 \times 10^{-3} \times 1000}{0.01} = 419$ mho $cm^2\ eq^{-1}$.

Illustration 27:

The conductivity of pure water in a conductivity cell with electrodes of cross-sectional area 4 cm^2 placed at a distance 2 cm apart is 8×10^{-7} S cm^{-1}. Calculate;

(a) The resistance of water.

(b) The current that would flow through the cell under the applied potential difference of 1 volt.

Solution:

Cell constant = $\frac{\ell}{a} = \frac{2}{4} = \frac{1}{2}$ cm^{-1}

(a) Also, $\kappa = \frac{1}{R} \times \frac{\ell}{a}$

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$$R = \frac{1}{\kappa} \times \frac{\ell}{a} = \frac{1}{8 \times 10^{-7}} \times \frac{1}{2} = 6.25 \times 10^5 \text{ ohm}$$

(b) From Ohm's law, $\frac{V}{i} = R$

$$\therefore \quad i = \frac{1}{6.25 \times 10^5} = 1.6 \times 10^{-6} \text{ ampere}$$

Illustration 28:

Molar conductance of 1 M solution of weak acid HA is 20 $ohm^{-1}\ cm^2\ mol^{-1}$. Find % dissociation of

$$HA : \begin{pmatrix} \Lambda_m^\circ(H^\oplus) = 350\ S\ cm^2\ mol^{-1} \\ \Lambda_m^\circ(A^\ominus) = 50\ S\ cm^2\ mol^{-1} \end{pmatrix}$$

Solution:

$$\Lambda_m^\circ(HA) = 350 + 50 = 400$$

$$\alpha = \frac{\Lambda_m^c}{\Lambda_m^\circ} \times 100 = \frac{20}{400} \times 100 = 5\ \%$$

Illustration 29:

Conductivity of an aqueous solution of 0.1 M HX (a weak mono-protic acid) is $5 \times 10^{-4}\ Sm^{-1}$.

Find pK_a[HX. Given : $\Lambda_m^\infty[H^\oplus] = 0.04\ Sm^2mol^{-1}$; $\Lambda_m^\infty[X^\ominus] = 0.01\ Sm^2mol^{-1}$

Solution:

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$$\underset{0.1(1-\alpha)}{HX} \rightleftharpoons \underset{0.1\alpha}{H^\oplus} + \underset{0.1\alpha}{X^\ominus}$$

$$\Lambda_m = \kappa \times \frac{1000}{C} \Rightarrow 5 \times 10^{-6} \times \frac{1000}{0.1} = 0.05\ \Omega^{-1}\ cm^2\ mol^{-1}$$

$$\alpha = \frac{\Lambda_m^c}{\Lambda_m^\infty} = \frac{0.05}{50} \Omega^{-1}cm^2mol^{-1} = 10^{-4}$$

$K_a = C\alpha^2 = 0.1 \times (10^{-4})^2 = 10^{-9}$

$pK_a = 9$

Illustration 30:

Specific conductance of 10^{-4} M n-Butyric acid aqueous solution is $1.9 \times 10^{-3}\ S\ m^{-1}$. If molar conductance of n-Butyric acid at infinite dilution is $380 \times 10^{-4}\ S\ m^2\ mol^{-1}$, then K_a for n-Butyric acid is :

Solution:

$$\Lambda_m = \frac{1.9 \times 10^{-3}}{1000 \times 10^{-4}} = 1.9 \times 10^{-2}$$

$$\alpha = \frac{1.9 \times 10^{-2}}{380 \times 10^{-4}} = 0.5$$

$$K_a = \frac{C\alpha^2}{1-\alpha} = \frac{10^{-4}(0.5)^2}{1-0.5} = 5 \times 10^{-5}\ M$$

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Illustration 31:

The specific conductance of a saturated AgCl solution is found to be 2.12×10^{-6} S cm^{-1} and that for water is 6×10^{-8} S cm^{-1}. The solubility of AgCl is : ($\lambda_{eq.}^{\infty}$ = 103 S equiv^{-1} cm^2)

Solution:

$\kappa_{Sol} = 2.12 \times 10^{-6}$ S cm^{-1}

$$S = \frac{(\kappa_{Ag\,sol.} - \kappa_{H_2O}) \times 1000}{(\Lambda_{eq.}^{0})_{AgCl}} = \frac{(212 \times 10^{-8} - 6 \times 10^{-8}) \times 1000}{103} = 2 \times 10^{-5}\,M$$

Illustration 32:

The value of Λ^{∞} for NH_4Cl, NaOH and NaCl are 129.8, 248.1 and 126.4 ohm^{-1} cm^2 mol^{-1} respectively. Calculate μ^{∞} for NH_4OH solution.

Solution:

$$\Lambda^{\infty}_{NH_4OH} = \Lambda^{\infty}_{NH_4Cl} + \Lambda^{\infty}_{NaOH} - \Lambda^{\infty}_{NaCl}$$

= 129.8 + 248.1 – 126.4

$\Lambda^{\infty}_{NH_4OH}$ = 251.5 ohm^{-1} cm^2 mol^{-1}

Illustration 33:

Calculate molar conductance for NH_4OH, given that molar conductance's for $Ba(OH)_2$, $BaCl_2$ and NH_4Cl are 523.28, 280.0 and 129.8 ohm^{-1} cm^2 mol^{-1} respectively.

Solution:

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$$\Lambda^{\infty}_{Ba(OH)_2} = \lambda^{\infty}_{Ba^{2\oplus}} + 2\lambda^{\infty}_{OH^{\ominus}} = 523.28 \quad \text{... (i)}$$

$$\Lambda^{\infty}_{BaCl_2} = \lambda^{\infty}_{Ba^{2\oplus}} + 2\lambda^{\infty}_{Cl^{\ominus}} = 280.00 \quad \text{... (ii)}$$

$$\Lambda^{\infty}_{NH_4Cl} = \lambda^{\infty}_{NH_4^{\oplus}} + \lambda^{\infty}_{Cl^{\ominus}} = 129.80 \quad \text{... (iii)}$$

$$\Lambda^{\infty}_{NH_4OH} = \lambda^{\infty}_{NH_4^{\oplus}} + \lambda^{\infty}_{OH^{\ominus}} = \ ?$$

$$\text{Eq.(iii)} + \frac{\text{Eq.(i)}}{2} - \frac{\text{Eq.(ii)}}{2} \text{ will gives}$$

$$\lambda^{\infty}_{NH_4^{\oplus}} + \lambda^{\infty}_{OH^{\ominus}} = \Lambda^{\infty}_{NH_4OH} = \frac{502.88}{2} = 251.44 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$$

Illustration 34:

The equivalent conductivities of acetic acid at 298 K at the concentrations of 0.1 M and 0.001 M are 5.20 and 49.2 S cm^2 eq.$^{-1}$ respectively. Calculate the degree of dissociation of acetic acid at these concentrations. Given that, $\lambda^{\infty}(H^{\oplus})$ and $\lambda^{\infty}(CH_3COO^{\ominus})$ are 349.8 and 40.9 ohm^{-1} cm^2 mol^{-1} respectively.

Solution:

Degree of dissociation is given by $\alpha = \frac{\lambda^c}{\lambda^{\infty}}$

(i) Evaluation of $\lambda^{\infty}_{CH_3COOH}$:

$$\lambda^{\infty}_{CH_3COOH} = \lambda^{\infty}_{CH_3COO^{\ominus}} + \lambda^{\infty}_{H^{\oplus}}$$

= 40.9 + 349.8 = 390.7 ohm^{-1} cm^2 eq.$^{-1}$

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(ii) Evaluation of degree of dissociation :

At C = 0.1 M $\alpha = \frac{\lambda^c}{\lambda^\infty} = \frac{5.20}{390.7} = 0.013$

i.e. 1.3%

At C = 0.001 M $\alpha = \frac{\lambda^c}{\lambda^\infty} = \frac{49.2}{390.7} = 0.126$

i.e. 12.6%

Illustration 35:

At infinite dilution the molar conductance of $Al^{3\oplus}$ and $SO_4^{2\ominus}$ ion are 189 and 160 Ω^{-1} cm^2 $mole^{-1}$ respectively. Calculate the equivalent and molar conductivity at infinite dilute of $Al_2(SO_4)_3$.

Solution:

$$\lambda^\infty_{eq.\left[Al_2(SO_4)_3\right]} = \frac{1}{3}\lambda^\infty_{Al^{3\oplus}} + \frac{1}{2}\lambda^\infty_{SO_4^{2\ominus}}$$

$$= \frac{1}{3}\times 189 + \frac{1}{2}\times 160$$

$= 143\ \Omega^{-1}\ cm^2\ eq^{-1}$

Molar conductivity = $\lambda_{eq} \times$ V. F. = 143 × 6

$= 858\ \Omega^{-1} cm^2\ mol^{-1}$

Illustration 36:

Find Λ_m^∞ (in $\Omega^{-1}\ cm^2\ mol^{-1}$) for strong electrolyte AB_2 in water at 25° from the following data.

Conc. C(mole/L)	0.25	1
$\Lambda_m(\Omega^{-1}\ cm^2/mol)$	160	150

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Solution:

$160 = \Lambda_m^\infty - b \times \sqrt{.25}$ ... (A)

$150 = \Lambda_m^\infty - b \times \sqrt{1}$ ... (B)

b = 20 and $\Lambda_m^\infty = 170$

y = 170 – 20x

Intercept = 170

$\Lambda_m^\infty = 170\ \Omega^{-1}\ cm^2\ mol^{-1}$

Illustration 37:

For any sparingly soluble salt $[M(NH_3)_4Br_2]\ H_2PO_2$

Given : $\lambda^0_{[M(NH_3)_4Br_2]^\oplus} = 400\ S.m^2.mol^{-1}$,

$\lambda^0_{H_2PO_2^\ominus} = 100\ S.m^2.mol^{-1}$

Specific resistance of saturated $[M(NH_3)_4Br_2]H_2PO_2$ solution is 200 Ω^{-1}m.

If solubility product constant of the above salt is 10^{-x}. What will be the value of x.

Solution:

$$\wedge_m^\infty = \kappa \times \frac{1000}{s} \times 10^{-6}$$

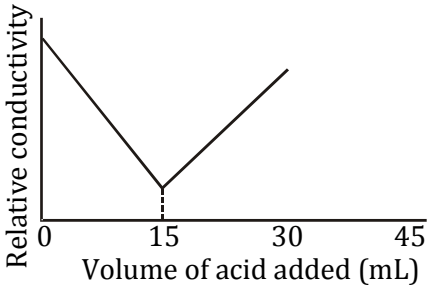
$$500 = \frac{1}{200} \times \frac{1000}{s} \times 10^{-6}$$

$S = 10^{-8}$ mol|L, $K_{sp} = S^2 = 10^{-16}$

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Illustration 38:

20 mL of KOH solution was titrated with 0.2 M H_2SO_4 solution in a conductivity cell. The data obtained were plotted to given the graph shown below :



the concentration of KOH solution was -

(A) 0.3 M (B) 0.15 M (C) 0.12 M (D) 0.075 M

Solution:

$20 \times M = 0.2 \times 2 \times 15 \quad \Rightarrow \quad M = 0.15$

Electrolysis :

The process of decomposition of an electrolyte by the passage of electricity is called **electrolysis** or electrolytic dissociation. It is carried out in electrolytic cell where electrical energy is converted into chemical energy. For electrolysis to take place two suitable electrodes are immersed in the liquid or solution of an electrolyte containing ions. When an electric potential is applied between the electrodes, the positively charged ions move towards the negative cathode and negatively charged ions move towards the positive anode, when a cation reaches the cathode, its takes up electron(s) and thus gets its charge neutralised. Thus the gain of electrons (decrease in oxidation number) means reduction takes place at the cathode.

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Similarly, when an anion reaches the anode, gives up electron(s) and thus gets discharged. Loss of electrons (Increase in oxidation number) means oxidation takes place at anode.

- The tendency of an electrode to lose electrons is known as the **oxidation potential**.
- The tendency of an electrode to gain electrons is known as the **reduction potential**.
- **Greater oxidation potential means stronger is the tendency to get oxidised and act as a reducing agent or reductant.**
- **Greater reduction potential means stronger is the tendency to get reduced and act as an oxidising agent (oxidant)**.

(a) Electrolysis of fused sodium chloride :

When fused sodium chloride is electrolysed, $Na^{\oplus}$ ions move towards the cathode and $Cl^{\ominus}$ ions move towards the anode. At cathode $Na^{\oplus}$ ions accept electrons to form sodium metal. At anode each $Cl^{\ominus}$ ion loses an electron to form Cl_2 gas.

- **At anode :** $Cl^{\ominus}(aq.) \longrightarrow \frac{1}{2} Cl_2 + e^{\ominus}$;
- **At cathode :** $Na^{\oplus}(aq.) + e^{\ominus} \longrightarrow Na(s)$;

(b) Electrolysis of aqueous solution of KBr

The solution of KBr contains $K^{\oplus}$, $Br^{\ominus}$ & small amounts of $H^{\oplus}$, $OH^{\ominus}$ (due to small dissociation of water)

- **At anode :** $2Br^{\ominus}(aq.) \longrightarrow Br_2(g) + 2e^{\ominus}$
- **At cathode :** $2H_2O(\ell) + 2e^{\ominus} \longrightarrow H_2(g) + 2OH^{\ominus}(aq)$

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- If more than one types of ions are present at a given electrode, then that ion is deposited which requires least energy. The energy required to liberate an ion is provided by the applied potential between electrodes. This potential is called **discharge or deposition potential**.

Note :

1. In aqueous solution most, electropositive metal cations.

   For eg. (s-block & $Al^{3\oplus}$) will not discharge at cathode instead H_2O is reduced.

   $\mathbf{2H_2O(\ell) + 2e^{\ominus} \longrightarrow H_2(g) + 2OH^{\ominus}(aq)}$

2. In aqueous solution cations of moderately electropositive metals (Mn, Co, Fe, Zn etc.) and least electropositive metals (Cu, Hg, Au, Ag, Pt) get discharged at cathode first.
3. **Active vs Inactive electrodes :**

- Sometimes the metal electrodes in the cell are active and the metals themselves are components of the half cell reactions or influence the reaction of electrode.
- For many redox reactions, however, there are no reactants or products capable of serving as electrodes. Inactive electrodes are used. For example rods of graphite or platinum, materials that conduct electrons into or out of the cell but does not take part in the half-reactions.
- Platinum is used ∵ highly conducting, unreactive.

  highly malleable and ductile.

Examples of electrolysis :

- **Electrolysis of aq. $PbBr_2$** (Using inert (Pt|graphite) electrodes).

  Cathode : $Pb^{2\oplus} + 2e^{\ominus} \rightarrow Pb(s)$ $\quad E^0 = -0.126V$

  Anode : $2Br^{\ominus} \rightarrow Br_2 + 2e^{\ominus}$ $\quad E^0 = -1.08\ V$

  $E^0_{cell} = -0.126 - (0.108) \times 10 = -1.206\ V$

  $E_{ext} > 1.206\ V$

- **Electrolysis of aq. $CuSO_4$** (Using inert (Pt|graphite) electrodes).

  Cathode : $Cu^{2\oplus} + 2e^{\ominus} \rightarrow Cu(s)$ $\quad E^0 = 0.34\ V$

  $2e^{\ominus} + 2H_2O(\ell) \rightarrow H_2(g) + 2OH^{\ominus}(aq)$ $\quad E^0 = -0.83V$

  Anode : $2SO_4^{2\ominus} \rightarrow S_2O_8^{2\ominus} + 2e^{\ominus}$ $\quad E^0 = -2.05\ V$

  $2H_2O(\ell) \rightarrow O_2 + 4H^{\oplus} + 4e^{\ominus}$ $\quad E^0 = -1.23\ V$

  ∴ Cu discharged at cathode and O_2 at anode.

- **Electrolysis of aq. NaCl** (Using inert (Pt|graphite) electrodes).

  Cathode : $Na^{\oplus} + e^{\ominus} \rightarrow Na$ $\quad E^0 = -2.71\ V$

  $2e^{\ominus} + 2H_2O(\ell) \rightarrow H_2(g) + 2OH^{\ominus}$ $\quad E^0 = -0.83\ V$

  Anode : $2Cl^{\ominus} \rightarrow Cl_2 + 2e^{\ominus}$ $\quad E^0_{OX} = -1.30\ V$

  $2H_2O(\ell) \rightarrow O_2 + 4H^{\oplus} + 4e^{\ominus}$ $\quad E^0_{OX} = -1.23\ V$

Rate of production of Cl_2 is more than rate of production of O_2 gas because of greater activation energy barrier for O_2 production, therefore Cl_2 is released at anode and H_2 at cathode.

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Note:

(i) As observed from electrode potential values discharge potential for O_2 is less than for Cl_2. According to thermodynamics, oxidation of H_2O to produce O_2 should take place on anode but experimentally (experiment from chemical kinetics) the rate of oxidation of water is found to be very slow. To increase it's rate, the greater potential difference has to be applied called over voltage or over potential.

Because of this, oxidation of $Cl^{\ominus}$ ions also become feasible and this takes place at anode. Electrode potential values do not take into account such effects as : Activation energy of process or non-uniform ionic concentration in solution.

(ii) Electrode potentials are thermodynamic intensive properties obtained experimentally under ideal & standard conditions. Sometimes in working conditions additional potentials are required for discharging. This difference is termed as overvoltage or overpotential.

Electrolysis using attackable (reactive) electrodes :

- **Electrolysis of aq. $CuSO_4$ using Cu electrode.**

Anode (oxidation) : $SO_4^{2\ominus} \rightarrow S_2O_8^{2\ominus} + 2e^{\ominus}$ $\quad E^0_{ox} = -2.05$ V

$2H_2O(\ell) \rightarrow O_2 + 4H^{\oplus} + 4e^{\ominus}$ $\quad E^0 = -1.23$ V

$Cu(s) \rightarrow Cu^{2\oplus} + 2e^{\ominus}$ $\quad E^0 = -0.34$ V

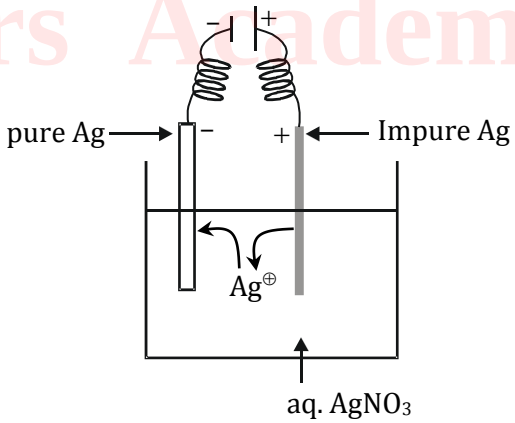
Cathode (reduction) : $Cu^{2\oplus} + 2e^{\ominus} \rightarrow Cu$ $\quad E^0 = +0.34$ V

$2H_2O(\ell) + 2e^{\ominus} \rightarrow H_2(g) + 2OH^{\ominus}$ $\quad E^0 = -0.83$ V

$\therefore$ Both oxidation and reduction of copper occurs and density of solution remains constant.

Electrolytic refining :

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Electrolysis of $AgNO_3$(aq) using Ag cathode & Ag anode.

Impure metal $\rightarrow$ Anode ;

Pure Metal $\rightarrow$ Cathode ;

Metal salt solution $\rightarrow$ Electrolyte

Anode : $NO_3^{\ominus} \rightarrow$ X (No reaction)

$2H_2O(\ell) \rightarrow O_2 + 4H^{\oplus} + 4e^{\ominus}$ $\quad E^0 = -1.23$V

$Ag(s) \rightarrow Ag^{\oplus}(aq) + e^{\ominus}$ $\quad E^0 = -0.80$V

(impure)

Cathode : $Ag^{\oplus} + e^{\ominus} \rightarrow Ag(s)$ (pure) $\quad E^0 = 0.8$V

$2H_2O(\ell) + 2e^{\ominus} \rightarrow H_2(g) + 2OH^{\ominus}$ $\quad E^0 = -0.83$V

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∴ Both oxidation and reduction of Ag occurs and mass transfer of Ag occurs from anode (impure Ag) to cathode (pure Ag). Electrical energy provided by battery is used for mass transfer of Ag from anode to cathode.

Products of electrolysis of some electrolytes :

S.No.	Electrolyte at anode	Electrode at cathode	Product obtained	Product obtained
(i)	Fused NaCl(molten)	Pt or Graphite	Cl_2	Na
(ii)	Aqueous NaCl(conc.)	Pt or Graphite	Cl_2	H_2
(iii)	dil.NaCl	Pt or Graphite	O_2	H_2
(iv)	Aqueous NaOH	Pt or Graphite	O_2	H_2
(v)	Fused NaOH	Pt or Graphite	O_2	Na
(vi)	Aqueous $CuSO_4$	Pt or Graphite	O_2	Cu
(vii)	Dilute HCl	Pt or Graphite	Cl_2	H_2
(viii)	Dilute H_2SO_4	Pt or Graphite	O_2	H_2
(ix)	Aqueous $AgNO_3$	Pt of Graphite	O_2	Ag
(x)	Aqueous CH_3COONa	Pt of Graphite	$CH_3–CH_3$ + CO_2	H_2

Faraday's laws of electrolysis :

Michael Faraday on basis of experiments deduced two important laws :

(a) Faraday's first law of electrolysis : This law states that **"The amount of a substance deposited or discharged at an electrode is directly proportional to the charge passing through the electrolytic solution".**

If a current of **I** amperes is passed for **t** seconds, (the quantity of charge Q in coulombs). If W gram of substances is deposited by Q coulombs of electricity, then

$$W \propto Q \propto i \times t$$

$$W = Z \times i \times t = \frac{E}{96500} \times i \times t = \eta \times \frac{E}{F} \times i \times t$$

moles of $e^{\ominus}$ = $n_e = \frac{\eta \times i \times t}{F}$ **= no. of equivalents of species discharged during electrolysis.**

Where 1 Faraday (1F) is defined as charge of 1 mole electrons = $eN_A = 1\ F \cong 96500\ C$

Hence faraday (F) is the quantity of charge in coulombs required to deposit one g equivalent of any substance.

E = Equivalent mass of species discharged

η = current efficiency in fraction if current efficiency is not mentioned, by default it is assumed to be 1 (100%).

Z is constant of proportionality and is known as **electrochemical equivalent**. Its value is different for different species, when Q = 1 coulomb , W = Z, thus electro chemical equivalent may be defined as the weight in grams of an element liberated by the passage of 1 coulomb of electricity.

Electrochemical equivalent of species (Z) = $\frac{E}{96500}$ gm | coulomb.

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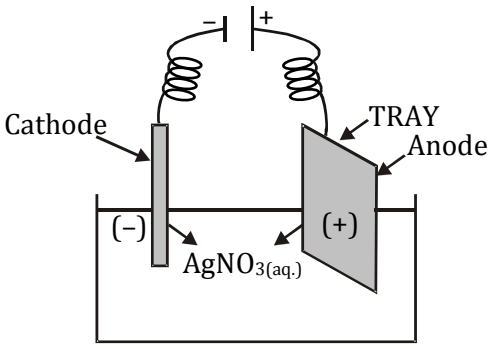
(b) Faraday's second law : This law states that the amounts of different substances deposited in different solutions connected in series at electrodes by passage of the same quantity of electricity are proportional to their equivalent masses(E).

$W \propto E$ (E = equivalent mass)

If W_1 and W_2 be the amounts of two different substances deposited at electrodes and E_1 and E_2 be the equivalent weights respectively then -

$$\frac{W_1}{W_2} = \frac{E_1}{E_2}$$

Application of electrolysis :



(i) Electroplating : Metal used for plating → ANODE
Object to be plated → CATHODE

(ii) Electrorefining : Impure metal → ANODE (see fig.)
Pure metal → CATHODE
Metal salt solution → electrolyte

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(iii) Electro-metallurgy : (Electrolytic reduction)

Ex. Electrolysis of NaCl (from seawater) Using Hg-cathode.

Anode : $2Cl^{\ominus} \longrightarrow Cl_{2(g)} + 2e$

Cathode : $Na^{\oplus} + e^{\ominus} \xrightarrow{Hg} Na\text{–}Hg$

Illustration 39:

How much charge is present on 1 mole of $Cu^{2\oplus}$ ion in faraday. (1 Faraday = 96500 coulomb)

Solution:

1 mole $e^{\ominus}$ = 1 mol proton = 1F

2 Faraday

Illustration 40:

For an element 'X' the process of oxidation is : $X_2O_4^{2\ominus} \longrightarrow$ product

If 965 A current when passed for 100 seconds for 0.1 mol of $X_2O_4^{2\ominus}$, find oxidation state of X in new compound ?

Solution:

$\overset{3\oplus}{X_2}\,O_4^{2\ominus} \longrightarrow 2X^{n\oplus}$

$\because$ oxidation process 3 < n, n-factor = 2 (n – 3)

so, $0.1 \times 2(n-3) = \frac{i \times t}{96500} = \frac{965 \times 100}{96500}$

$2(n-3) = 10$, $n - 3 = 5$, $n = 8$

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Illustration 41:

How many litres of chlorine at 1atm & 273K will be deposited by 100 amp. current flowing for 5 hours through molten NaCl?

Solution:

$Q = It = 100 \times 5 \times 60 \times 60 = 18 \times 10^5 C$

$W = ZQ = \frac{E}{96500} \times 18 \times 10^5 = \frac{18E}{96500} \times 10^5 = 662.2\ g, \left(E_{Cl_2} = \frac{71}{2} = 35.5\right)$

$\because$ Volume of 71 g Cl_2 at 1atm & 273K = 22.4 L

$\therefore$ Volume of 662.2 g Cl_2 at NTP = $\frac{22.4}{71} \times 662.2 = 208.9$ L

Illustration 42:

How much time is required for complete decomposition of two moles of water at anode using 4 amperes current ?

Solution:

$H_2O(l) \longrightarrow H_2(g) + \frac{1}{2}O_2\ (g)$

For H_2 : $2 \times 2 = \frac{4 \times t}{96500} \Rightarrow t = 96500$ sec,

Illustration 43:

How much charge (in F) must flow through solution during electrolysis of aq. Na_2SO_4 at 0ºC and 1 atm to produce 33.6 L of product gases at 50% current efficiency ?

Solution:

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$$H_2O \xrightarrow{2F} H_2 + \frac{1}{2}O_2$$

$$V \qquad V|2$$

$\Rightarrow \frac{3}{2}V = 33.6\,L$

$\Rightarrow V = 22.4$ L

$Q \times \frac{50}{100} = 2F \Rightarrow Q = 4F$

Illustration 44:

A Solution of copper (II) sulphate is electrolysed between copper electrodes by a current of 10.0 amperes passing for one hour. What changes occur at the electrodes and in the solution?

Solution:

According to Faraday's first law of electrolysis :

The reaction at cathode :

$$Cu^{2\oplus} + 2e^{\ominus} \longrightarrow Cu$$

$$63.5 \qquad 2 \times 96500\ C$$

The quantity of charge passed = I × t = (10 amp) × (60 × 60s) = 36000 C.

2 × 96500 C of charge deposit copper = 63.5 g

36500 C of charge deposit copper = $\frac{(63.5\,g)}{(2 \times 96500\,C)} \times (36000\,C) = 11.84\,g$

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Thus, 11.84 g of copper will dissolve from the anode and the same amount from the solution will get deposited on the cathode. The concentration of the solution will remain unchanged.

Illustration 45:

An aqueous solution of Na_2SO_4 was electrolysed for 10 min. 82 ml of a gas was produced at anode and collected over water at 27°C at a total pressure of 580 torr. Determine the current that was used in amp. , Given : Vapour pressure of H_2O at 27°C = 10 torr, R = 0.082 atm L|mol-K

Solution:

(1.6A)

At anode reaction will be

$2H_2O \rightarrow O_2 + 4H^{\oplus} + 4e^{\ominus}$

Vo_2 collected = 82 ml

By PV = nRT

$$\Rightarrow \frac{(580-10)}{760}\times\frac{82}{1000}=(n_{o_2})\times 0.0821\times 300,\ n_{o_2}=\frac{1}{400}$$

By Faraday law $\frac{W}{E}=\frac{i\times t}{96500}$

$$\left(\frac{W}{M}\right)\times n=\frac{i\times t}{96500},$$

$$\left(\frac{1}{400}\times 4\right)=\frac{i\times 10\times 60}{96500}\Rightarrow i = 1.6\ A$$

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Illustration 46:

The same current if passed through solution of silver nitrate and cupric salt connected in series. If the weight of silver deposited is 1.08 g. calculate the weight of copper deposited.

Solution:

According to faradays second law

$$\frac{W_1}{W_2}=\frac{E_1}{E_2}\Rightarrow\frac{1.08}{W_2}=\frac{108}{31.75}\Rightarrow W_2 = 0.3175\ g$$

Some Commercial Batteries :

Any battery or cell that we use as a source of electrical energy is basically an electrochemical cell where oxidising and reducing agents are made to react by using a suitable device. In principle, any redox reaction can be used as the basis of an electrochemical cell, but there are limitations to the use of most reactions as the basis of practical batteries. A battery should be reasonably right and compact and its voltage should not vary appreciably during the use.

There are mainly two types of cells :

(i) primary cells and (ii) secondary cells. In primary cells, the reaction occurs only once and the battery then becomes dead over a period of time and cannot be used again. (For example, dry cell, mercury cell.) Contrary to this, secondary cells can be recharged by passing a current through them so that they can be used again and again. (For example, lead storage battery, nickel- cadmium storage cell.)

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Primary Batteries :

Dry cell or Laclanche cell :

The most familiar type of battery is the dry cell which is a compact of Leclanche cell known after its discoverer Leclanche (fig.) : In this cell, the anode consists of a zinc container and the cathode is a graphite rod surrounded by powdered MnO_2 and carbon. The space between the electrodes is filled with a moist paste of NH_4Cl and $ZnCl_2$. The electrode reactions are complex, but they can be written approximately as follows.

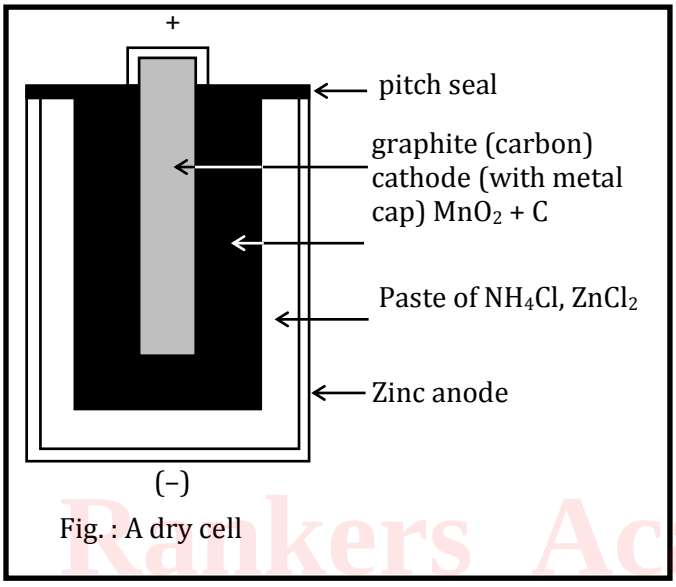
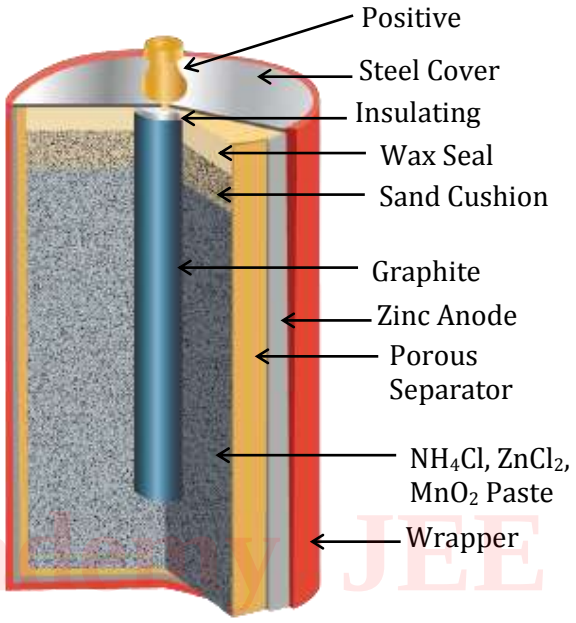


Fig. : A dry cell



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- **Anode** $Zn(s) \longrightarrow Zn^{2\oplus} + 2e^{\ominus}$
- **Cathode** $MnO_2 + NH_4^{\oplus} + e^{\ominus} \longrightarrow MnO(OH) + NH_3$

In the cathode reaction, manganese is reduced from the +4 oxidation state to the +3 state. Ammonia is not liberated as a gas but combines with $Zn^{2\oplus}$ to form $Zn(NH_3)_4^{2\oplus}$ ion. The cell has a potential of nearly 1.5 V.

Mercury cell :

Mercury cell, suitable for low current devices like hearing aids, watches, etc. consists of zinc & mercury amalgam as anode and a paste of HgO and carbon as the cathode. The electrolyte is a paste of KOH and ZnO. The electrode reactions for the cell are given below:

Anode : $Zn(Hg) + 2OH^{\ominus} \rightarrow ZnO(s) + H_2O + 2e^{\ominus}$

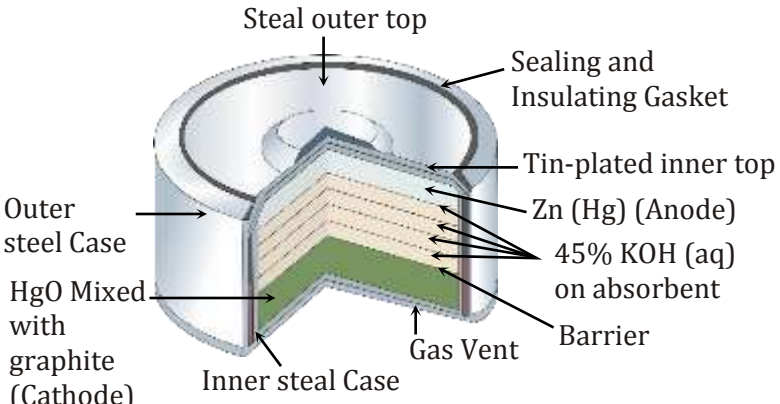
Cathode : $HgO + 2H_2O + 2e^{\ominus} \rightarrow Hg(l) + 2OH^{\ominus}$

The overall reaction is represented by

$Zn(Hg) + HgO(s) \rightarrow ZnO(s) + Hg(l)$

The cell potential is approximately 1.35 V and remains constant during its life as the overall reaction does not involve any ion in solution whose concentration can change during its life time.

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Commonly used mercury cell.
The reducing agent is zinc and the oxidizing agent is mercury (II) oxide.

Secondary Batteries :

A secondary cell after use can be recharged by passing current through it in the opposite direction so that it can be used again. A good secondary cell can undergo a large number of discharging and charging cycles.

Lead storage cell :

The most important secondary cell is the lead storage battery commonly used in automobiles and invertors. It consists of a lead anode and a grid of lead packed with lead dioxide (PbO_2) as cathode. A 38% solution of sulphuric acid is used as an electrolyte.

The cell reactions when the battery is in use are given below:

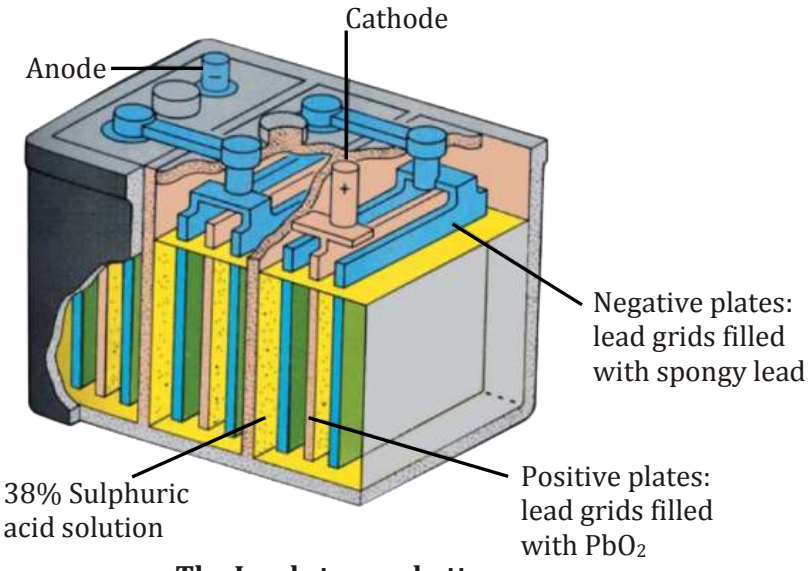
Anode : $Pb(s) + SO_4^{2\ominus}(aq) \rightarrow PbSO_4(s) + 2e^{\ominus}$

Cathode : $PbO_2(s) + SO_4^{2\ominus}(aq) + 4H^{\oplus}(aq) + 2e^{\ominus} \rightarrow PbSO_4(s) + 2H_2O(\ell)$

i.e., overall cell reaction consisting of cathode and anode reactions is:

$Pb(s) + PbO_2(s) + 2H_2SO_4(aq) \rightarrow 2PbSO_4(s) + 2H_2O(I)$

On charging the battery the reaction is reversed and $PbSO_4(s)$ on anode and cathode is converted into Pb and PbO_2, respectively.



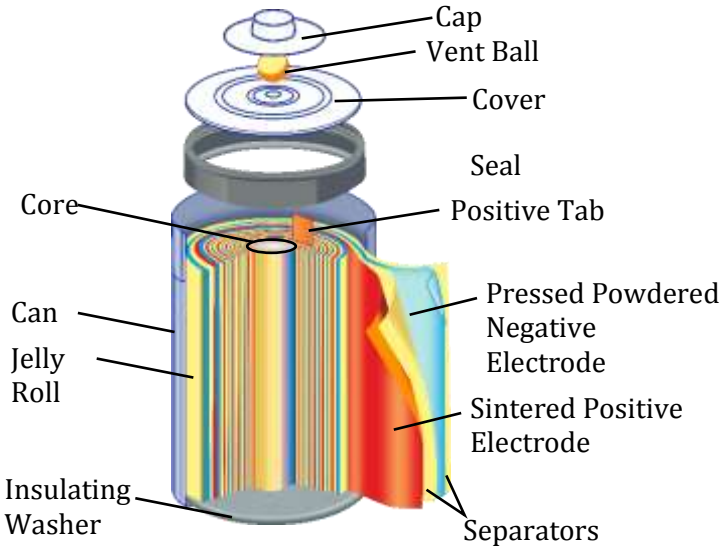
The Lead storage battery

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Nickel-cadmium cell :

Another important secondary cell is the nickel-cadmium cell which has longer life than the lead storage cell but more expensive to manufacture. We shall not go into details of working of the cell and the electrode reactions during charging and discharging. The overall reaction during discharge is:

$Cd\,(s) + 2Ni(OH)_3(s) \rightarrow CdO(s) + 2Ni(OH)_2\,(s) + H_2O(l)$



A rechargeable nickel-cadmium cell in a jelly roll arrangement and separated by a layer soaked in moist sodium or potassium hydroxide.

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Fuel cells :

Production of electricity by thermal plants is not a very efficient method and is a major source of pollution. In such plants, the chemical energy (heat of combustion) of fossil fuels (coal, gas or oil) is first used for converting water into high pressure steam. This is then used to run a turbine to produce electricity. We know that a galvanic cell directly converts chemical energy into electricity and is highly efficient. It is now possible to make such cells in which reactants are fed continuously to the electrodes and products are removed continuously from the electrolyte compartment. Galvanic cells that are designed to convert the energy of combustion of fuels like hydrogen, methane, methanol, etc. directly into electrical energy are called fuel cells. One of the most successful fuel cells uses the reaction of hydrogen with oxygen to form water. The cell was used for providing electrical power in the Apollo space programme. The water vapours produced during the reaction were condensed and added to the drinking water supply for the astronauts. In the cell, hydrogen and oxygen are bubbled through porous carbon electrodes into concentrated aqueous sodium hydroxide solution. Catalysts like finely divided platinum or palladium metal are incorporated into the electrodes for increasing the rate of electrode reactions. The electrode reactions are given below:

Cathode : $O_2(g) + 2H_2O(I) + 4e^{\ominus} \rightarrow 4OH^{\ominus}(aq)$

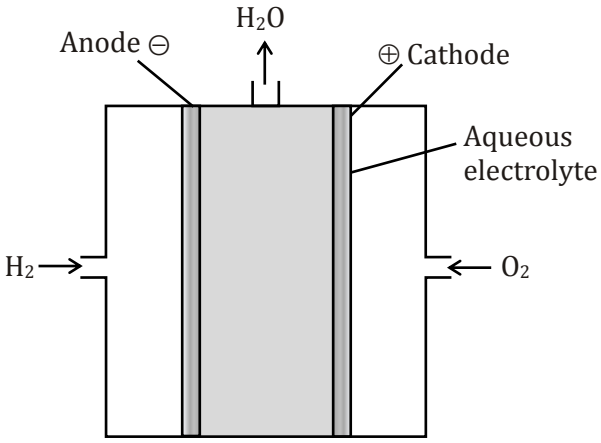
Anode : $2H_2(g) + 4OH^{\ominus}(aq) \rightarrow 4H_2O(I) + 4e^{\ominus}$

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Overall reaction being :

$2H_2(g) + O_2(g) \rightarrow 2H_2O(I)$

The cell runs continuously as long as the reactions are supplied Fuel cells produce electricity with an efficiency of about 70% compared to thermal plants whose efficiency is about 40%. There has been tremendous progress in the development of new electrode materials, better catalysts and electrolytes for increasing the efficiency of fuel cells. These have been used in automobiles on an experimental basis. Fuel cells are pollution free and in view of their future importance, a variety of fuel cells have been fabricated and tried.

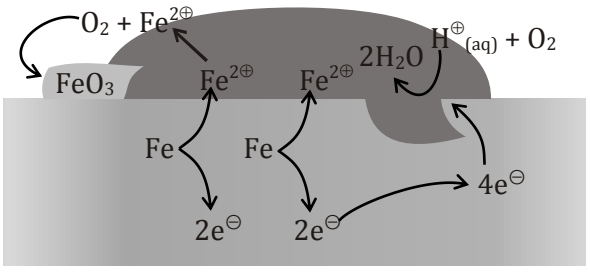


Fuel cell using H_2 and O_2 produces electricity

Corrosion :

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Corrosion slowly coats the surfaces of metallic objects with oxides or other salts of the metal. The rusting of iron, tarnishing of silver, development of green coating on copper and bronze are some of the examples of corrosion. It causes enormous damage to buildings, bridges, ships and to all objects made of metals especially that of iron. We lose crores of rupees every year on account of corrosion. In corrosion, a metal is oxidised by loss of electrons to oxygen and formation of oxides. Corrosion of iron (commonly known as rusting) occurs in presence of water and air. The chemistry of corrosion is quite complex but it may be considered essentially as an electrochemical phenomenon. At a particular spot of an object made of iron, oxidation takes place and that spot behaves as anode and we can write the reaction



Oxidation: $Fe(s) \rightarrow Fe^{2\oplus}(aq) + 2e^{\ominus}$

Reduction: $O_2(g) + 4H^{\oplus}(aq) + 4e^{\ominus} \rightarrow 2H_2O(I)$

Atmospheric

Corrosion of iron in atmosphere

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Anode : $2Fe(s) \longrightarrow 2Fe^{2\oplus} + 4e^{\ominus}$

$E^{\ominus}_{(Fe^{2\oplus}/Fe)} = -0.44\ V$

Electrons released at anodic spot move through the metal and go to another spot on the metal and reduce oxygen in presence of $H^{\oplus}$ (which is believed to be available from H_2CO_3 formed due to dissolution of carbon dioxide from air into water. Hydrogen ion in water may also be available due to dissolution of other acidic oxides from the atmosphere). This spot behaves as cathode with the reaction

Cathode: $O_2(g) + 4H^{\oplus}(aq) + 4e^{\ominus} \longrightarrow 2H_2O\ (I)$ $\qquad E_{H^{\oplus}|O_2|H_2O} = 1.23\ V$

The overall reaction being:

$2Fe(s) + O_2(g) + 4H^{\oplus}(aq) \longrightarrow 2Fe^{2\oplus}(aq) + 2H_2O(I)$ $\qquad E^{\ominus}_{(Cell)} = 1.67\ V$

The ferrous ions are further oxidised by atmospheric oxygen to ferric ions which come out as rust in the form of hydrated ferric oxide ($Fe_2O_3 \cdot xH_2O$) and with further production of hydrogen ions.

Prevention of corrosion is of prime importance. It not only saves money but also helps in preventing accidents such as a bridge collapse or failure of a key component due to corrosion. One of the simplest methods of preventing corrosion is to prevent the surface of the metallic object to come in contact with atmoshphere. This can be done by covering the surface with paint or by some chemicals (e.g. bisphenol). Another simple method is to cover the surface by other metals (Sn, Zn, etc.) that are inert or react to save the object. An electrochemical method is to provide a sacrificial electrode of another metal (like Mg, Zn, etc.). Which corrodes itself but saves the object.

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